IGECNet: An Interaction-Guided Multimodal Fusion Network With Elevation Constraint for Remote Sensing Semantic Segmentation

Haoxue Zhang, Gang Xie, *Member, IEEE*, Linjuan Li, Chenhao Chang, Xinlin Xie, Jinchang Ren, *Senior Member, IEEE*, and Heng Li

***Abstract*—Semantic segmentation of multimodal remote sensing images plays a crucial role in geospatial analysis. However, existing methods often struggle to effectively integrate spatial–spectral and geometric–geographic information, capture multiscale contextual features, and preserve the fine-grained boundaries of geo-objects. To address these challenges, we propose an interaction-guided multimodal fusion network with elevation constraint, referred to as IGECNet. The proposed network comprises four key components. First, a multimodal correlation interaction (MCI) mechanism adaptively fuses features from high-resolution remote sensing images (HRRSIs) and digital surface models (DSMs) using multiscale dilated convolutions and cross-attention mechanisms, thereby enhancing complementary relationships while preserving modality-specific characteristics. Second, a channel reorder enhancement module (CREM) hierarchically decomposes channel-wise features based on entropy and global average pooling values, improving spatial–spectral discriminability. Third, a frequency-guided context decoder (FGCD) refines segmentation outputs by leveraging global semantic context via the Fourier transform and fine boundary details through local feature fusion. In addition, a new loss function, jointly constrained by semantic, boundary, and elevation consistency, enforces elevation consistency from DSMs gradients and boundary alignment. Extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, and DroneDeploy segmentation benchmarks demonstrate that IGECNet achieves state-of-the-art performance. The source code will be publicly available upon publication.**

***Index Terms*—Cross-scale integration, domain knowledge loss function, frequency domain guidance decoder, multimodal fusion, remote sensing semantic segmentation.**

NOMENCLATURE

F_H	Features extracted by the HRRSI encoder.
F_D	Features extracted by the DSM encoder.
$F_{H_{\text{dc}}}$	Features after dilated convolution in the MCI-HRRSI branch.
$F_{D_{\text{dc}}}$	Features after dilated convolution in the MCI-DSM branch.
$F_{H_{\text{mul}}}$	Cross-interaction HRRSI features in the MCI-HRRSI branch.
$F_{D_{\text{mul}}}$	Cross-interaction DSM features in the MCI-DSM branch.
F_{fuH}	Fused features of the MCI-HRRSI branch.
F_{fuD}	Fused features of the MCI-DSM branch.
Q, K, V	Query, key, value.
Q_h, K_h	Horizontal components of Q and K.
K_h^{T}	Transpose of K_h.
S_h	Attention score matrix for height dimension.
Attn	Attention maps over both dimensions.
$\text{Attn}_h, \text{Attn}_h^{\text{T}}$	Horizontal attention map and its transpose.
CA_h, CA_w	Cross-attention outputs.
CA_{mid}	Intermediate cross-attention feature.
$\text{CA}_{\text{fuH}}, \text{CA}_{\text{fuD}}$	Final fused HRRSI/DSM features after attention.
A_c, E_c, G_c	Attribute vector, entropy, and global average pooling of channel c.
$\mathcal{H}(\cdot)$	Histogram function.
x_i^c, x_j^c	Pixel value at histogram bins of channel c.
$x_{(i,j)}^c$	Pixel value at position (i, j).
p_i	Normalized histogram-bin probability.
$\mathcal{C}_k$	Channel index set of cluster k.

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f_k	Cluster-wise feature representation.
f_{Enh}	Multiscale dilation-enhanced feature.
X_f, Y_f	FFT-domain input/output features.
X, Y	Spatial-domain features (before FFT and after IFFT).
$y_i^c, \widehat{y_i^c}$	Ground truth and predicted probability.
$\mathcal{L}_{\text{ce}}, \mathcal{L}_{\text{ce}}^{\text{mid}}, \mathcal{L}_{\text{edge}}, \mathcal{L}_{\text{elev}}$	Semantic consistency, mid-level CE, edge preservation, and elevation decision losses.
f_L	Intermediate layer feature maps.
y_i^M	Mismatched pixels.
$S_{\text{mean}}^c, S_{\text{mode}}^c, S_p^c$	Mean, mode, and proxy scores of class c.
conv_k^d	Dilated convolution with kernel k and dilation d.
$\text{conv}_q, \text{conv}_k, \text{conv}_v$	1×1 projections for Q/K/V.
$\times$	Matrix multiplication.
$\odot$	Element-wise multiplication.
cat(·)	Concatenation.
softmax(·)	Softmax activation.
CBR(·)	Conv + BN + activation block.
BN(·)	Batch normalization.
SiLU(·)	Sigmoid linear unit.
ReLU(·)	Rectified linear unit.
Cluster(·)	Clustering operation.
$\mathcal{F}_{2D}(\cdot)$, $\mathcal{F}_{2D}^{-1}(\cdot)$	FFT and inverse FFT transforms.
Sobel(·)	Sobel edge extraction.
MSE(·)	Mean squared error.

I. Introduction

SEMANTIC segmentation is a fundamental task in remote sensing, enabling pixel-wise classification of images to extract detailed spatial and structural information. This capability is essential for applications such as precision agriculture [1], natural resource management [2], and disaster assessment [3], where accurate land-cover classification supports informed decision-making. High-resolution imagery plays a critical role in achieving accurate semantic segmentation, as its finer spatial details notably enhance the model's ability to discriminate complex geographic features. Despite these advantages, high-resolution remote sensing images (HRRSIs) suffer from inherent challenges such as spectral noise, shadows, occlusions, and intraclass heterogeneity, which significantly complicate segmentation tasks. Recent advancements in satellite and airborne remote sensing technologies have significantly improved the availability and quality of high-resolution data, further driving research into techniques designed to fully exploit the capabilities of high-resolution imagery.

Traditional segmentation methods [4], [5], [6] rely on handcrafted descriptors that are labor-intensive and often lack robustness. Deep learning has advanced HRRSIs' semantic segmentation by enabling automated, data-driven feature extraction [7], [8], [9], [10], [11]. However, single-modality methods remain limited compared to multimodal approaches, which integrate optical imagery with complementary sources refers to the integration of optical imagery with complementary modalities such as digital surface models (DSMs) [12], [13], [14], synthetic aperture radar (SAR) [15], [16], [17], or light detection and ranging (LiDAR) [18], [19], [20]. Such auxiliary data enrich 3-D structural cues, reduce sensitivity to shadows and occlusions [21], and enhance model generalizability [22]. Despite these advantages, multimodal segmentation still faces several challenges, which are structurally independent but hierarchically progressive across the processing pipeline as follows.

1) *Complexity of Heterogeneous Data Integration:* Effectively integrating spatial, spectral, and elevation data from diverse sources remains a significant challenge. Advanced fusion techniques are needed to combine complementary features from multiple modalities while appropriately weighing their contributions to achieve optimal segmentation performance.
2) *Challenges in Mining and Utilizing Multiscale Features:* HRRSIs contain features at diverse spatial scales, making it essential to accurately extract and integrate multiscale information to segment objects of varying sizes. Many existing methods lack the ability to combine global contextual cues with fine-grained local details, resulting in suboptimal performance—especially for complex or small objects.
3) *Insufficient Edge Preservation and Domain Knowledge Utilization:* Accurately identifying and preserving edge information is particularly challenging for closely adjacent objects or regions with subtle differences. Many current methods do not incorporate edge-guided mechanisms, leading to blurred or inaccurate boundaries. Furthermore, limited incorporation of domain knowledge—such as prior terrain information—hampers segmentation precision and applicability in remote sensing contexts.

These three challenges are structurally independent yet hierarchically progressive: modality heterogeneity must first be resolved to enable effective fusion, which subsequently facilitates refined semantic abstraction, ultimately supporting topology-consistent boundary decoding. Various methods have been proposed for fusing multimodal images at different levels: data-level [23], feature-level [14], [24], and decision-level [25] fusions.

Data-level fusion typically involves linking images from multiple modalities, such as utilizing ancillary data as additional channels for HRRSIs [26] or performing preliminary fusion after basic processing [23]. However, this approach often fails to effectively extract and utilize the complementary information inherent in multimodal data, leading to suboptimal segmentation performance.

Decision-level fusion, similar to ensemble learning, involves combining the final outputs or decisions from different networks. For instance, SoftFormer [25] uses a joint-key learning module to mitigate feature redundancy and derives the final result from three branch amalgamations. The Ebbinghaus-curve-guided multisource remote sensing classification network [27] employs λ, β, and ϵ as weight coefficients obtained through adaptive learning to enhance global

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decision-level fusion. However, its effectiveness depends heavily on the robustness of the individual classifiers and the chosen fusion strategy. If the individual models are not sufficiently robust or fail to capture complementary information across modalities, the fusion may yield only marginal improvements. In addition, the performance of this approach is strongly influenced by the choice of fusion strategy—whether it be voting, averaging, or more complex weighted combinations. Inappropriate fusion rules can result in information loss or redundancy, limiting the potential performance gains in multimodal segmentation.

Researchers have increasingly focused on feature-level fusion, where features extracted by encoders from different modalities are integrated through a fusion module before entering the segmentation head for per-pixel classification. Compared with data-level fusion, this approach more effectively uses complementary information by leveraging learned features from each modality. Feature-level fusion can be further categorized into two distinct paradigms based on how the fused features are utilized. The first paradigm, single-stage fusion, is widely adopted in existing methods. In this design, fused features are used exclusively at one stage of the pipeline, either (1) to enhance encoder feature extraction or (2) to guide the decoder during segmentation. For instance, FtransUNet [14] employs shallow-level feature fusion modules and a vision transformer to learn representative cross-modal features, which are then passed to the decoder. CMGFNet [13] introduces a gated fusion module to adaptively learn discriminative features, using residual-like depth-wise separable convolutions in the decoder to upsample features from both RGB and cross-modal streams. CIMFNet [28] proposes a cross-layer gate fusion module to integrate multiscale complementary information across modalities. The PACSCNet [29] encodes multiscale information into a unified layer by learning feature differences between adjacent layers within each modality. While effective in certain scenarios, single-stage fusion often underutilizes multimodal information, as the fused features are restricted to a single application stage.

The second paradigm, two-stage hierarchical fusion, fundamentally differs by leveraging fused features at both encoder and decoder stages simultaneously. Compared with single-stage methods, this design provides three major advantages as follows.

1) *Progressive Cross-Modal Representation Learning:* By injecting fused features back into the encoder, the network continuously refines low- and mid-level representations under multimodal constraints. This alleviates modality dominance (e.g., RGB overshadowing DSM) and strengthens the mutual complementarity of modalities across scales.

2) *Enhanced Semantic Abstraction With Optimization Stability:* Reusing fused features in the decoder not only enriches high-level semantic reasoning but also provides a regularization effect, stabilizing the optimization trajectory by constraining both encoder and decoder with consistent multimodal cues. This dual guidance mitigates the risk of overfitting to modality-specific noise or redundancy.

3) *Topology-Aware Structural Consistency:* The hierarchical utilization allows boundary-level details and elevation priors to propagate through the decoding process, ensuring that object contours and class boundaries are refined in harmony with global semantics. This addresses one of the main shortcomings of single-stage fusion, where boundary artifacts and semantic-structural inconsistencies often persist. For example, the multimodal fusion network [24] fuses multimodal features and forward weighted branch features to the next-level encoders, with predictions obtained through progressive upsampling. CEGFNet [30] incorporates multiple gate fusion modules within a dual-stream backbone to extract complementary features from both modalities. Local-to-global cross modal attention aware fusion [31] adopts a feature enhancement module and a feature interaction and fusion module in each encoder stage, allowing each branch to be optimized for the unique characteristics of its modality. Despite these advances, research on two-stage architectures remains relatively limited, and existing designs often underexploit how to systematically integrate physical priors or optimize cross-modal interactions across scales, leaving substantial room for further development.

In addition to multimodal fusion strategies, boundary-aware optimization has also been extensively studied for remote sensing semantic segmentation. Recent works have introduced boundary refinement modules [32], [33], [34], edge-guided attention [35], [36], and explicit edge supervision [10], [37] to improve delineation of adjacent objects. While these approaches enhance edge consistency, most are designed for single-modality RGB images [35], [36], [37], [38], [39], and lack consideration of multimodal correlations or elevation priors.

Building upon these observations, we propose an interaction-guided multimodal fusion network with an elevation constraint, referred to as IGECNet, specifically designed for multimodal remote sensing semantic segmentation. Unlike prior works that apply fused features only once, IGECNet propagates cross-modal correlations into both encoder and decoder stages, forming a holistic two-stage pipeline. This design directly addresses the aforementioned challenges by introducing robust cross-attention to balance heterogeneous inputs, enhancing multiscale feature aggregation through entropy- and frequency-guided modules, and incorporating elevation priors to sharpen boundaries and enforce elevation consistency. Specifically, IGECNet leverages intermodal correlations to strengthen feature representations through the following coordinated components. The multimodal correlation interaction (MCI) mechanism is designed to capture both global context and local details across modalities, enabling effective and robust feature fusion. The channel reorder enhancement module (CREM) and the frequency-guided context decoder (FGCD) collaboratively address challenges related to multiscale feature integration and spatial awareness, thereby improving segmentation accuracy, particularly for small or complex objects. Furthermore, the semantic-boundary-elevation (SBE) joint constraint loss, guided by domain knowledge, improves boundary detection and aligns segmentation outputs with elevation features, effectively preserving edges and enhancing elevation consistency.

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Collectively, these modules form a comprehensive framework that substantially enhances the precision and robustness of multimodal semantic segmentation. The primary contributions of this study are summarized as follows.

1) We propose IGECNet, a multimodal segmentation framework that propagates feature interactions into both encoder and decoder stages. Unlike conventional single-stage fusion, IGECNet establishes a hierarchical two-stage pipeline that enables continuous cross-modal alignment, thereby strengthening feature representation and improving segmentation accuracy.
2) The MCI module employs a dual-fusion strategy that couples cross-modal attention with dilated convolutions, specifically tailored for HRRSI-DSM integration. This design simultaneously captures complementary information and adapts to scale variations, offering a more flexible and robust fusion process than prior attention- or convolution-based approaches.
3) The CREM module dynamically reorganizes channel responses to enhance multiscale integration, while the FGCD introduces a frequency-guided fusion mechanism that combines global semantics with local details. Together, they provide a novel spatial-frequency coordination strategy that improves the segmentation of small, heterogeneous, or elevation-sensitive objects.
4) The SBE joint constraint loss unifies three complementary priors: semantic consistency, boundary refinement, and elevation alignment. By explicitly embedding elevation cues, it achieves topology-aware boundary preservation and enhances model generalization in complex environments.

The remainder of this article is organized as follows. Section II presents the IGECNet methodology. Section III details the experiments conducted, including ablation studies and comparisons with existing methods. Section IV discusses the fusion strategies and limitations of the proposed network. Finally, Section V summarizes the conclusions.

II. Methodology

A. Motivation and Overall Architecture

Multimodal HRRSI-DSM segmentation remains affected by modality imbalance, inconsistent multiscale feature representation, and insufficient utilization of elevation cues for boundary preservation. To address these limitations, a hierarchical two-stage fusion framework is designed, which departs from conventional practices that typically perform multimodal interaction only at a single decoding stage [14], [40], [41] or rely on shallow residual coupling that cannot regulate cross-modal semantics throughout the feature hierarchy. Rather than introducing novelty at the level of individual operators, the proposed framework reorganizes interaction pathways, information flow, and multimodal reasoning into a unified architectural design tailored to the demands of HRRSI-DSM segmentation. Specifically, the framework incorporates: 1) a two-stage bidirectional interaction pipeline, in which dual-branch multimodal correlation modules progressively propagate modality-aware refinements during feature encoding; 2) an entropy-guided dynamic channel reordering mechanism together with a spatial-frequency cooperative decoder, enabling structured multiscale coordination rather than isolated feature fusion operations; and 3) an SBE joint constraint that explicitly embeds terrain-related priors into the learning objective to enhance topology-aware prediction. Through this hierarchical design, modality-specific semantics are progressively aligned, structurally meaningful cues are preserved, and refined cross-modal interactions are consistently propagated across feature scales.

Based on these design principles, the overall architecture of IGECNet is organized as illustrated in Fig. 1. The proposed IGECNet adopts a multistage dual-stream architecture designed to effectively integrate and process HRRSIs and DSMs inputs. The network consists of two independent encoders, multiple cross-modal fusion mechanisms, a channel reordering module, and a decoder equipped with frequency-guided context aggregation, with an additional domain knowledge guided loss to refine structural accuracy.

The network begins with two independent encoders extracting modality-specific features from the HRRSIs and DSMs. Each encoder extracts features at four spatial resolutions, progressively reducing the resolution by a factor of two at each stage while expanding the channel dimensions as $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. This design ensures that both fine-grained local cues and high-level semantic information are captured. At every encoder stage, a pair of MCI modules is invoked. Specifically, MCI-HRRSI uses HRRSIs' features as the primary branch while integrating complementary cues from DSM features, whereas MCI-DSM symmetrically takes DSMs as the primary branch. This bidirectional fusion strategy enables mutual reinforcement between modalities at each resolution level. The outputs from MCI-HRRSI and MCI-DSM are then concatenated and processed by the CREM, which reorganizes channel-wise responses to emphasize discriminative semantics and suppress noise, particularly in modality-inconsistent regions. The fused and refined representations are passed into the decoder, which incorporates FGCD blocks at each stage. FGCD combines high-level semantic features from deeper layers with spatially detailed cues from shallower features through lateral connections and progressive upsampling. In addition to conventional spatial domain fusion, FGCD introduces frequency domain guidance, which enhances multiscale feature utilization and improves spatial awareness, especially for small or structurally complex objects. Finally, the SBE loss is applied at the output stage. It unifies semantic consistency, boundary refinement, and elevation alignment into a single objective, enforcing topology-aware supervision. By embedding morphological and elevation priors, the SBE loss strengthens object boundary delineation and improves robustness in heterogeneous environments.

B. Multimodal Correlation Interaction

Multimodal features are difficult to integrate due to the distinct data distribution of each modality. Existing fusion strategies typically perform cross-modal interaction only through shallow residual coupling or at a single decoding stage, which leaves most encoder layers blind to modality

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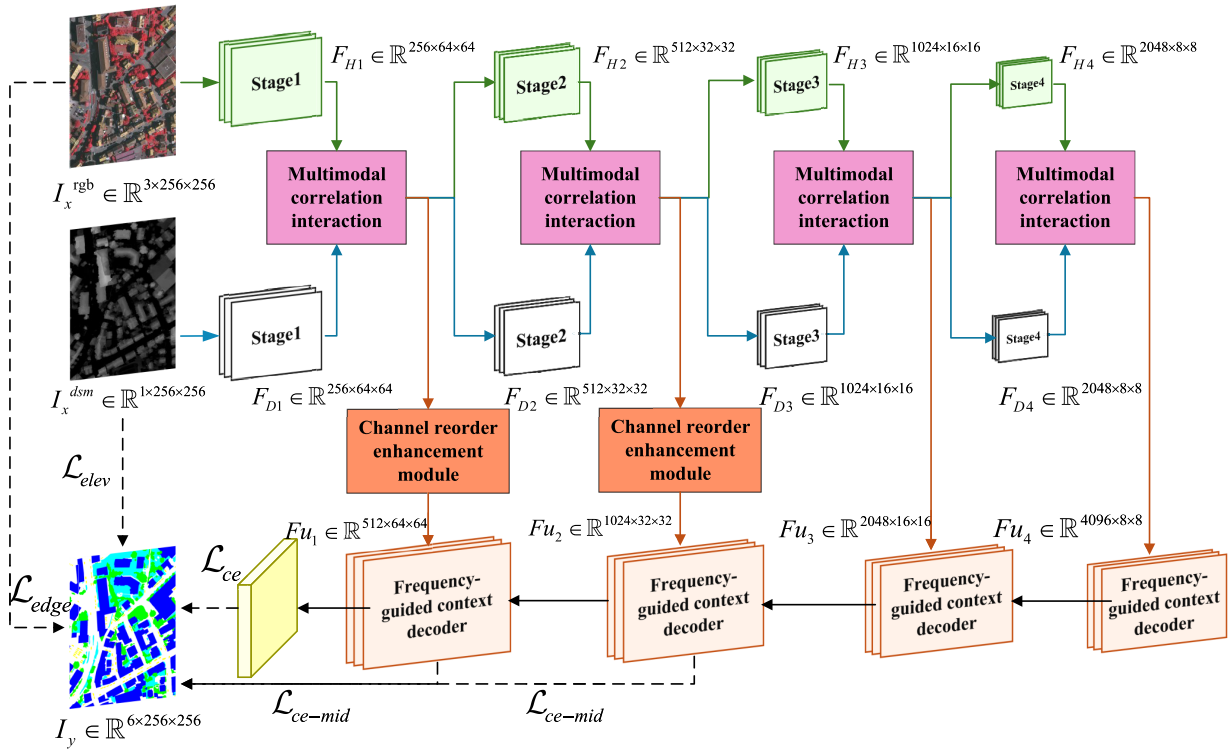


Fig. 1. Overall framework of proposed network. The network follows a two-stage design that integrates hierarchical feature extraction, cross-modal fusion (MCI), and adaptive decoding to leverage spatial (CREM and FGCD) and elevation information (SBE loss) effectively to enhance segmentation accuracy.

discrepancies. As a result, these approaches are unable to regulate modality-specific semantics consistently throughout the feature hierarchy, cannot effectively correct early stage imbalances, and fail to capture the recursive dependencies that accumulate across scales. To overcome these limitations, the proposed MCI introduces a layerwise, dual-branch, bidirectional interaction mechanism, allowing the fused representation from one modality to directly influence the feature extraction process of the other across all resolution levels. This progressive mechanism reorganizes the information flow so that cross-modal cues are not only exchanged but also reinjected into the encoding process, enabling continuous refinement across hierarchical stages. In contrast to conventional attention- or convolution-based fusion that operates as an isolated module, MCI functions as an interaction pipeline, in recursively allied. This integrated design yields a depth-aware multimodal reasoning process, which emerges from the hierarchical interaction pipeline, rather than from any individual operator such as cross-attention or dilated convolution. In addition, through this progressive bidirectional interaction, MCI allows the HRRSI and DSM streams to iteratively refine one another across feature levels, enabling a degree of modality alignment and hierarchical correlation modeling that cannot be achieved when cross-modal interaction is limited to a single encoder or decoder stage.

As shown in Fig. 1, the MCI is incorporated into a Res2Net encoder. And the distinct branches dedicated to handling the two modalities (MCI-HRRSI and MCI-DSM) ensure that both feature streams are effectively coupled and utilized. With F_H and F_D denoting the features extracted by the HRRSIs and DSMs encoder branches, respectively, Fig. 2 shows that the fused feature F_{fu_H}, which guides the HRRSIs encoder branch to the next learning stage, is

$$\begin{aligned} F_{H_{\mathrm{dc}}} &= \mathrm{cat}\left(\mathrm{conv}_k^d\left(F_H\right)\right) \\ F_{H_{\mathrm{mul}}} &= \mathrm{cat}\left(F_{H_{\mathrm{dc}}} \odot F_D, F_H\right) \\ F_{\mathrm{fu}_H} &= \mathrm{SiLU}\left(\mathrm{BN}\left(\mathrm{conv}\left(F_{H_{\mathrm{mul}}}\right)\right)\right) \end{aligned} \tag{1}$$

where conv_k^d denotes dilated convolution, with kernel size k and dilation rate d. We use four parallel convolutions with a kernel size of 3 $\times$ 3 and dilation rates of 1, 2, 4, and 8, enabling the network to capture multiscale contextual information by aggregating features at different receptive field sizes. $F_{H_{\mathrm{dc}}}$ and $F_{H_{\mathrm{mul}}}$ are intermediate features resulting from the dilated convolutions and fusion, respectively. All remaining variables and operators are summarized in Nomenclature. The dilation rates are varied to reduce the parameters and mitigate overfitting while capturing multilevel local features through multiscale dilation. The fused feature F_{fu_D}, which guides the DSMs encoder branch, is

$$\begin{aligned} F_{D_{\mathrm{dc}}} &= \mathrm{cat}\left(\mathrm{conv}_k^d\left(F_D\right)\right) \\ F_{D_{\mathrm{mul}}} &= \mathrm{cat}\left(F_{D_{\mathrm{dc}}} \odot F_H, F_D\right) \\ F_{\mathrm{fu}_D} &= \mathrm{SiLU}\left(\mathrm{BN}\left(\mathrm{conv}\left(F_{D_{\mathrm{mul}}}\right)\right)\right). \end{aligned} \tag{2}$$

The feature maps $F_{H_{\mathrm{dc}}}$ and $F_{D_{\mathrm{dc}}}$ are intermediate representations and are further enhanced by cross modal feature interactions based on the cross-attention mechanism [42], as shown in Fig. 2. This facilitates the transfer and integration of complementary information between modalities, enriching the feature interactions between the HRRSIs and DSMs streams.

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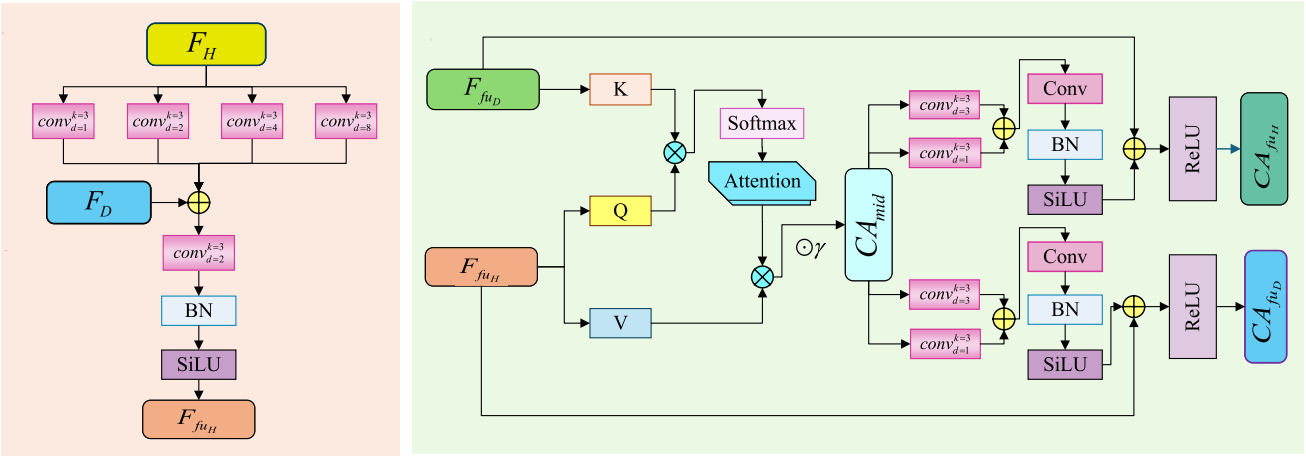


Fig. 2. Details of the proposed MCI. (a) Multiscale receptive field extraction using dilated convolutions with multiple dilation rates, which captures contextual information while reducing parameter complexity and mitigating overfitting. (b) Cross-attention mechanism for integrating complementary information between modalities, enhancing inter-stream interaction.

Cross-attention captures long-range dependencies and contextual relationships between the HRRSIs and DSMs features, addressing the failure of CNNs to fully utilize long-range spatial information. The cross-attention mechanism features dual attention, operating along the height and width dimensions to create a comprehensively fused feature map. The process begins by concatenating the feature maps from both modalities and passing them through a series of convolutions to produce query (Q), key (K), and value (V) tensors for the attention mechanism

$$\begin{aligned} Q &= \text{conv}_q\left(F_{\text{fu}_D}\right) \\ K &= \text{conv}_k\left(F_{\text{fu}_H}\right) \\ V &= \text{conv}_v\left(F_{\text{fu}_H}\right) \end{aligned} \tag{3}$$

where conv_q, conv_k, and conv_v are 1×1 convolutions that project high-dimensional feature maps into lower dimensional spaces. This dimensionality reduction is crucial for computational efficiency and enables the attention mechanism to capture spatial dependencies effectively across both the height and width dimensions [43]. The spatial correlations along the height dimension are

$$S_h = Q_h \times K_h^{\text{T}} + \infty \tag{4}$$

where S_h is the score matrix for the height dimension, and ∞ is a large negative value used to mask irrelevant regions. The spatial correlations along the width dimension are computed in the same manner, yielding S_w from Q_h and K_h. Masking ensures that attention focuses only on meaningful areas of the feature map, thereby improving model robustness and accuracy. The attention maps for both dimensions are then obtained using the softmax function

$$\text{Attn} = \text{softmax}\left(\text{cat}\left(S_h, S_w\right)\right). \tag{5}$$

The outputs of the height and width attention mechanisms are computed in the same manner, where CA_h is

$$\text{CA}_h = V_h \times \text{Attn}_h^{\text{T}}. \tag{6}$$

Finally, the outputs of the height and width attention mechanisms are combined and scaled by a learnable parameter γ and then added to the fused feature maps $\text{CA}_{\text{fu}H}$ and $\text{CA}_{\text{fu}D}$ from the HRRSIs and DSMs branches, respectively,

$$\begin{aligned} \text{CA}_{\text{mid}} &= \gamma \odot \left(\text{CA}_h + \text{CA}_w\right) \\ \text{CA}_{\text{fu}H} &= \text{ReLU}\left(\text{CBR}\left(\text{cat}\left(\text{conv}_d^k\left(\text{CA}_{\text{mid}}\right)\right)\right) + F_{\text{fu}_H}\right) \\ \text{CA}_{\text{fu}D} &= \text{ReLU}\left(\text{CBR}\left(\text{cat}\left(\text{conv}_d^k\left(\text{CA}_{\text{mid}}\right)\right)\right) + F_{\text{fu}_D}\right) \end{aligned} \tag{7}$$

where $\text{CBR}(\cdot)$ is a CBR (convolution, batch norm, and activation) block combining convolution, batch normalization, and activation. This operation integrates the attention-enhanced features with the original fused feature maps, refining the final outputs for both the HRRSIs and DSMs encoder branches.

Through this progressive bidirectional interaction, MCI enables the HRRSI and DSM streams to iteratively refine one another across feature hierarchies, achieving more stable modality alignment and enhanced hierarchical correlation modeling than fusion strategies that restrict cross-modal interaction to a single stage or shallow coupling.

C. Channel Reorder Enhancement Module

Remote sensing features often exhibit pronounced inter-channel variability and scale-dependent semantic contribution, which makes simple channel concatenation, uniform attention weighting, or fixed convolutional integration insufficient for achieving balanced multimodal representation. Existing approaches typically treat channels as unordered feature vectors and fuse them through permutation-invariant operations (e.g., concatenate, channel attention, and static convolution mixing), which do not explicitly distinguish redundant channels from structurally informative ones. Consequently, these methods are unable to regulate channel-wise information distribution prior to multiscale fusion and tend to propagate channel-level redundancy and modality-specific bias into the decoder. To address these limitations, the CREM is introduced,

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as shown in Fig. 3, which performs attribute-guided channel reorganization followed by structured multiscale enrichment. Unlike prior designs that merely apply scalar attention weighting or fixed reindexing, CREM adopts an information-theoretic criterion to reorganize channels before multiscale processing, ensuring that channels with higher entropy and stronger global discriminability dominate subsequent fusion.

The module consists of two stages: 1) channel clustering, which groups channels based on intrinsic attributes and performs a semantic-guided reorganization rather than a mere permutation, and 2) multiscale enhancement, which enriches contextual information within each group. In channel clustering, each channel of the feature map is evaluated based on entropy [44], [45], and global average pooling; these metrics quantify the channel information richness and global contribution, promoting effective representation. These attributes form a 2-D vector for clustering channels with similar characteristics using the K-means algorithm [46]. For balanced representation, the clustered channels are adjusted to uniform dimensions using 1×1 convolution. Specifically, given a feature map $X \in R^{C\times H\times W}$ with C channels, an attribute vector $A_c \in R^2$ is derived for each channel c as

$$A_c = \begin{bmatrix} E_c \\ G_c \end{bmatrix} \tag{8}$$

where the entropy E_c of the cth channel is

$$E_c = -\sum_{i=1}^{256} p_i \log(p_i), \quad p_i = \frac{\mathcal{H}\left(x_i^c\right)}{\sum_{j=1}^{256} \mathcal{H}\left(x_j^c\right)} \tag{9}$$

where x^c denotes the pixel values in the cth channel, p_i represents the normalized histogram bin probabilities, and $\mathcal{H}(\cdot)$ is the histogram of an image x_c on a specific channel i. The global average pooling value G_c of the cth channel is

$$G_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{(i,j)}^c \tag{10}$$

where H and W are the height and width of the feature map, respectively, and $x_{(i,j)}^c$ is the pixel value at location (i, j). The above attributes are combined into an attribute vector for clustering. Then, the K-means clustering algorithm [46] is used to partition the channels into K clusters based on their attribute vectors

$$\mathcal{C}_k = c|\text{Cluster}(A_c) = k, \quad k = 1, 2, \ldots, K. \tag{11}$$

For each cluster $\mathcal{C}_k$, the grouped feature maps are aggregated and processed further. For balanced representation, the channel dimensions of each cluster are adjusted to a target size T using 1×1 convolution

$$f_k = \text{Conv}_{1\times 1}\left(X_{\mathcal{C}_k}\right), \quad f_k \in R^{\text{T}\times H\times W}. \tag{12}$$

It is worth noting that this process constitutes a weighted recombination of channels guided by information-theoretic attributes, ensuring that discriminative patterns are preserved and redundant responses are suppressed, rather than a permutation of indices.

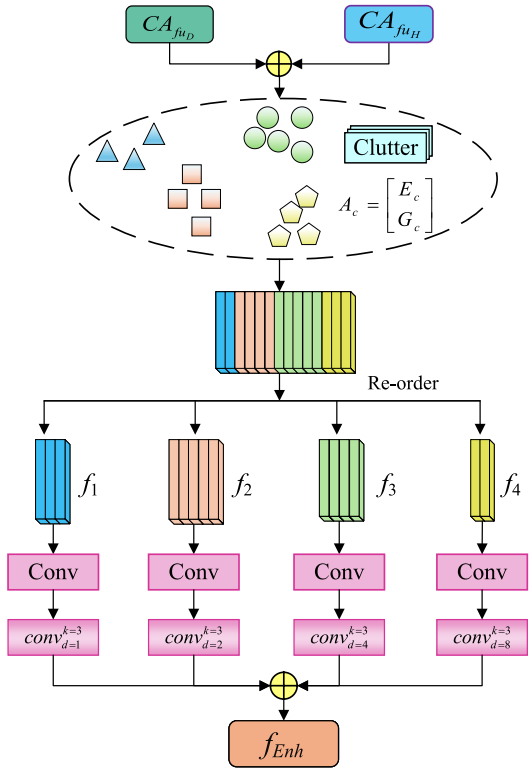


Fig. 3. Structure of CREM. This module optimizes the feature distribution and enhances multiscale contextual representation using a process with two stages: channel clustering and multiscale enhancement. The first stage groups feature maps based on their intrinsic attributes, and the second enriches the contextual information within each group, effectively addressing the high variability and multiresolution characteristics of remote sensing images.

In multiscale enhancement, each cluster undergoes dilated convolutions with varying dilation rates, capturing both fine-grained details and broader contextual information. The outputs from the different dilation scales are concatenated to form the enriched representation

$$f_{\text{Enh}} = \text{cat}\left(\text{Conv}_{d=1}(f_k), \text{Conv}_{d=2}(f_k) \right. \\ \left. \text{Conv}_{d=4}(f_k), \text{Conv}_{d=8}(f_k)\right) \tag{13}$$

where f_k denotes features extracted by convolution at a dilation rate $d = r$ and $f_{\text{Enh}} \in \mathbb{R}^{4\text{T}\times H\times W}$. Other dimension information is shown in Table X.

This two-stage channel reorganization and enrich design differs from conventional multiscale blocks by explicitly reorganizing channel ordering prior to multiscale processing, which reduces the propagation of redundant responses and ensures that informative channels exert a dominant influence during spatial contextual aggregation. Through this attribute-guided channel reallocation and multidilation refinement, CREM provides a semantically ordered and noise-reduced feature basis for the subsequent spatial-frequency decoding stage in FGCD.

D. Frequency-Guided Context Decoder

Decoding multimodal remote sensing features requires simultaneously preserving high-frequency boundary information, maintaining long-range semantic consistency, and

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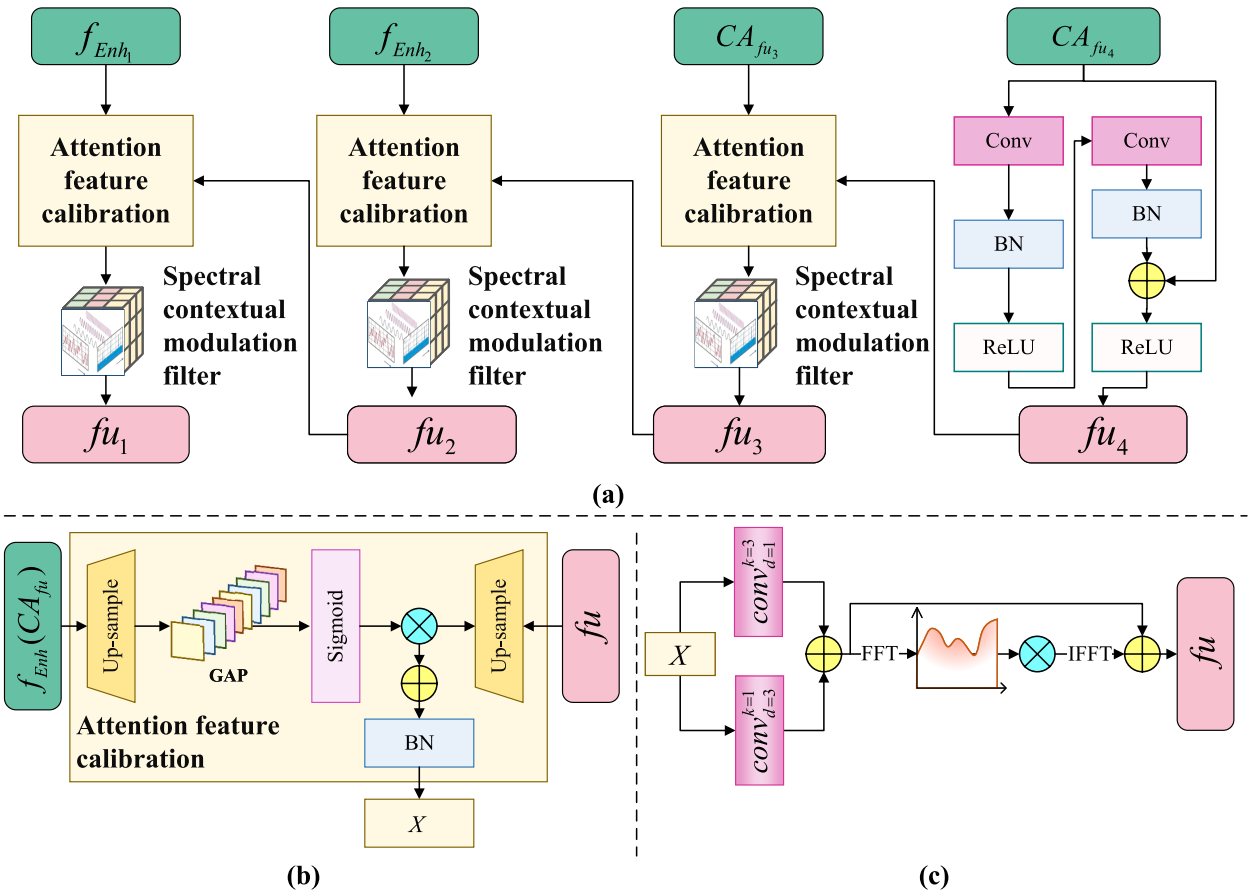


Fig. 4. (a) Structure of FGCD. (b) AFC module. This module integrates spatial- and frequency domain features via global-scale fusion to strengthen local and global contextual information. (c) SCMF. This component, which is the core of the FGCD, operates in the frequency domain with learnable complex-valued filters to modulate the spectral components adaptively, thereby refining the spatial representations and enhancing the segmentation quality.

preventing the loss of elevation-related structural cues. To address these demands, the FGCD is introduced, a design that organizes spatial-spectral processing within a unified decoding strategy. Instead of applying frequency operations as isolated plug-in components, FGCD integrates spatial reasoning, spectral modulation, and global contextual fusion within a single decoding pipeline, thereby providing a structured pathway for multimodal feature refinement.

FGCD consists of two complementary components: the attention feature calibration (AFC) module and the spectral contextual modulation filter (SCMF), as shown in Fig. 4. AFC provides global-scale spatial-spectral feature fusion, whereas SCMF offers adaptive complex-valued spectral modulation, which reshapes feature representations prior to their projection back into the spatial domain. Through their joint operation, these components establish a task-driven spatial-frequency interaction mechanism that enhances structural fidelity and semantic discrimination beyond what conventional decoders achieve through purely spatial convolutions or frequency-domain operations applied in isolation. The decoding process begins with the AFC module upsampling the input features and performing global-scale fusion on the frequency-enhanced and spatial domain features. This is achieved by integrating the output of frequency domain feature modulation with the original spatial domain feature map, followed by multiplicative interaction to enhance both local and global contextual representations. The refined feature map is passed through a batch normalization layer to ensure numerical stability and further improve feature distribution.

Next, the SCMF modulates the feature responses in the frequency domain. Initialized as a complex-valued parameter tensor with dimensions corresponding to the spatial and channel dimensions of the input, the SCMF functions as a learnable frequency domain filter. These complex weights are optimized during training to adjust the frequency components of the input features adaptively. Specifically, the feature map is transformed from the spatial domain into the frequency domain via 2-D fast Fourier transform (FFT) [47], [48], and the learned filters are applied elementwise to the frequency domain. Consequently, the model selectively emphasizes informative spectral components, thereby refining the spatial structure of the decoded features and enhancing semantic segmentation precision. Unlike a conventional feed-forward network (FFN) that applies linear transformations in the spatial domain without explicit frequency modeling, the SCMF performs modulation directly in the spectral domain. This allows the

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model to explicitly capture global dependencies via low-frequency components and to refine fine-grained boundaries via high-frequency cues. As a result, the SCMF provides more discriminative and adaptive feature enhancement than a simple FFN, thereby improving multimodal feature fusion and segmentation performance.

The frequency domain transformation is

$$X_f = \mathcal{F}_{2-D}(X), \; Y_f = X_f \odot W \tag{14}$$

where $\mathcal{F}_{2-D}(\odot)$ is the FFT operation, X_f is the frequency domain representation of the input X, and W is a complex-valued weight tensor whose real and imaginary components are jointly initialized using a zero-mean Gaussian distribution with a standard deviation of 0.02. After filtering, the feature map is transformed back into the spatial domain using inverse 2-D FFT

$$Y = \mathcal{F}_{2-D}^{-1}\left(Y_f\right) \tag{15}$$

where $\mathcal{F}_{2-D}^{-1}(\odot)$ is the inverse FFT (IFFT) operation, and all remaining notations are defined in Nomenclature. After frequency domain processing, the model better captures long-range dependencies and high-frequency details, which may be difficult to capture using traditional convolutional operations. The SCMF enhances feature representation by capturing both fine-grained, high-frequency details, such as edges, and large-scale contextual information through global frequency processing. Rather than functioning as an isolated spectral filter, SCMF is embedded within the decoding process, so that its frequency-domain adjustments directly influence multiscale upsampling, spatial refinement, and global contextual aggregation. This integration enables the decoder to reinforce boundary responses, preserve elevation-related structural discontinuities, and maintain long-range semantics consistency in heterogeneous scenes. Furthermore, the spectral modulation emphasizes informative frequency components, complementing spatial operations in capturing large-scale structural patterns and fine-grained object boundaries.

The decoder structure also incorporates a CBR block and global-scale fusion. The SCMF refines the upsampled features by multiplying them with the fused output of the frequency domain feature modulation and the original feature map. This process enhances the local and global contexts within the feature map, which then undergoes batch normalization for further refinement.

In summary, the FGCD provides a unified spatial-frequency decoding strategy in which spectral modulation and spatial refinement are jointly integrated, rather than serving as independent functional operators. This design constitutes a structural contribution at the decoding stage, enhances semantic precision, boundary fidelity, and elevation-sensitive feature reconstruction, and thereby supports more detailed feature representation in complex remote sensing scenarios for improved semantic segmentation.

E. SBE Joint Constraint Loss Function

Accurate multimodal remote sensing segmentation requires not only pixel-wise semantic consistency, but also structural fidelity and explicit alignment with elevation-related priors inherent to HRRSI-DSM data. To meet these demands, the SBE joint constraint is introduced, formulated as a unified learning objective that combines semantic supervision, boundary preservation, and terrain-aware regularization within a single framework. Rather than relying on independent loss terms, the proposed design incorporates domain-specific knowledge into the optimization objective, ensuring that semantic predictions, structural boundaries, and DSM-derived terrain cues are jointly enforced during model training. This loss function consists of three main terms: cross-entropy loss for the predicted and intermediate segmentation maps, edge preservation loss for boundary refinement, and elevation decision loss for incorporating DSMs' terrain information. Each term in this comprehensive loss function addresses a specific aspect of segmentation, ensuring that the model effectively integrates multiscale information and learns discriminative features, preserves accurate object boundaries, and aligns with terrain characteristics, ultimately improving segmentation accuracy.

1) Semantic Consistency Loss: The cross-entropy loss, which measures the discrepancy between the ground-truth (GT) and predicted segmentation masks, is the primary loss function for pixel-wise classification and is calculated as

$$\mathcal{L}_{\text{ce}} = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_i^c \log\left(\widehat{y_i^c}\right) \tag{16}$$

where N is the number of pixels; C is the number of classes; and y_i^c and $\widehat{y_i^c}$ are the GT and predicted probabilities, respectively, for class c at pixel i. This loss term, which guides the model to minimize the pixel-wise classification error, is needed to learn the segmentation map. In addition to the final prediction, we introduce the intermediate cross-entropy loss to provide auxiliary supervision in different encoder stages. This term helps the network perform consistently across the feature extraction layers, particularly in multiscale feature extraction. The intermediate cross-entropy loss is computed for the intermediate feature maps f_k as

$$\mathcal{L}_{\text{ce}}^{\text{mid}} = \sum_{k=1}^{K} \mathcal{L}_{\text{ce}}\left(f_k, y_i^c\right). \tag{17}$$

This provides multiscale semantic regularization and stabilizes feature learning across network depths, particularly under conditions of modality imbalance.

2) Edge Preservation Loss: Edges should be preserved to detect boundaries accurately, especially in complex remote sensing data, where objects with similar spectral characteristics may have defined boundaries. We propose the edge preservation loss to refine the segmentation output, considering boundary details. This loss is computed using Sobel edge detection to extract edges from the predicted segmentation maps, followed by the calculation of the mean squared error (mse) between the GT and predicted edges

$$\mathcal{L}_{\text{edge}} = \frac{1}{N}\sum_{i=1}^{N} \text{mse}\left(\text{Sobel}_{\hat{\text{y}}_\text{i}}, \text{Sobel}_{\text{y}_\text{i}}\right) \tag{18}$$

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where $\text{Sobel}_{\hat{y}_i}$ and Sobel_{y_i} are the predicted and GT edge maps for pixel i, respectively. The edge loss emphasizes the preservation of fine-grained object boundaries.

This term emphasizes structurally meaningful gradient responses, improving the delineation of small-scale and heterogeneous objects commonly encountered in high-resolution remote sensing imagery.

3) Elevation Decision Loss: We incorporate terrain-specific knowledge into the segmentation process using the elevation decision loss, which integrates DSM's terrain features to guide the model's segmentation decisions, ensuring consistency between predicted and elevation characteristics. The elevation decision loss computes dynamic proxies per class based on class statistics (mean and mode values) and adjusts the class proxies based on the elevation data to improve terrain consistency. First, we calculate the mean S_{mean} and mode S_{mode} values per class using the class statistics. These proxies represent the adaptive elevation values per class, aiming to capture terrain-specific patterns during segmentation. Then, we compute class proxies S_p^c to compute the DSM differences while adjusting for the elevation features. Finally, the elevation decision loss is calculated by comparing the predicted and true class labels using the proxy-based elevation consistency

$$\mathcal{L}_{\text{elev}} = \frac{1}{\sum_1^N (p_i - 1)^2 \times y_i^M} \tag{19}$$

where p_i is the probability of pixel i, y_i^M represents the mismatched pixels of predictions, and the loss is normalized using the number of valid pixels. Specifically, the proxy S_p^c is based on the mean difference between classes, which determines the proximity of classes in elevation as

$$S_p^c = \lambda \cdot S_{\text{mean}}^c + (1 - \lambda) \cdot S_{\text{mode}}^c \tag{20}$$

where λ is computed based on the minimum mean difference between classes.

This design encourages elevation-consistent predictions, assisting the model in resolving ambiguous or spectrally similar classes by leveraging DSM-derived terrain-related priors.

4) SBE Joint Constraint Function: The domain-knowledge-guided loss function integrates multiple terms to improve semantic segmentation in remote sensing. By combining the cross-entropy, edge preservation, and elevation decision losses, we address the key challenges of multiscale feature integration, boundary preservation, and elevation consistency. The total loss function is

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{ce}}^{\text{mid}} \times \alpha + \mathcal{L}_{\text{edge}} \times \beta + \mathcal{L}_{\text{elev}} \times \theta \tag{21}$$

where α, β, and θ are weighted factors used to prioritize each term during training.

In summary, the SBE joint constraint serves as a terrain-aware regularization mechanism that promotes multiscale semantic consistency, reinforces structurally meaningful boundaries, and guides the optimization process with DSM-derived elevation cues, enabling topology-aware, elevation-consistent, and boundary-precise segmentation in complex HRRSI-DSM scenarios.

III. Experiments

Experiments are conducted on several multimodal land-use classification benchmark datasets to evaluate the effectiveness of IGECNet. We select a representative HRRSIs and DSMs as the primary data sources for evaluation.

A. Datasets

1) ISPRS Vaihingen Dataset: The Vaihingen dataset comprises 33 images with a ground sampling distance (GSD) of 0.09 m. The image sizes range from 3816 × 2550 pixels to 2555 × 1388 pixels, collectively covering a total area of 1.38 km^2 in Vaihingen, Germany. Each image contains three spectral bands: infrared (IR), red (R), and green (G). The dataset is annotated with six categories: car, tree, low. veg. (low vegetation/grass), building, impervious surface ("imp. suf."), and clutter. During training, the images are divided into 256 × 256 patches without overlap. The training/testing split follows reference [14], with 12 images for training and four for testing. All image patches are normalized to [0, 1] and augmented with random flips and rotations.

2) ISPRS Potsdam Dataset: This dataset includes 38 high-resolution true orthophoto (TOP) images paired with corresponding DSMs, each measuring 6000 × 6000 pixels with a GSD of 0.05 m. These TOP images comprise four channels: R, G, blue (B), and IR. They are annotated with the same six categories in the Vaihingen dataset, ensuring consistency in classification. IR-RG false-color composites serve as the experimental HRRSIs. We extract 256 × 256 patches without overlap and follow the common split of 18 images for training and six for testing. The same normalization and augmentation strategies are applied.

3) DroneDeploy Segmentation Dataset: This dataset consists of various drone-captured aerial scenes, with each scene featuring a GSD of 0.1 m. The optical images have an average size of 4864 × 3648 pixels. The labels are supplied as PNG files, with seven distinct colors representing seven categories (building, clutter, vegetation, water, ground, car, and ignore), ensuring precise class differentiation. The dataset is randomly partitioned into training, validation, and testing sets with a 7:2:1 ratio. During training, 512 × 512 patches are cropped from each scene. Input images are normalized to the [0, 1] range, and color jittering is applied to improve generalization.

B. Evaluation Metrics

The effectiveness of the proposed method is comprehensively evaluated using three widely adopted metrics: intersection over union (IoU), overall accuracy (OA), and $F1$-score. These metrics collectively enable a multifaceted evaluation of segmentation from both global and class-specific perspectives. They are formulated as

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{22}$$

$$F1 - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{23}$$

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TABLE I
ABLATION EXPERIMENTS ON PROPOSED MODULES ON VAIHINGEN DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA

Methods	IoU/(%)					mIoU/(%)	OA/(%)	F1
	Imp. Suf.	Building	Low. Veg.	Tree	Car			
Baseline	82.94	91.16	63.26	80.78	67.96	77.22	89.50	0.7230
Baseline + MCI	85.13	93.06	64.46	81.55	75.37	79.91	90.50	0.7376
Baseline + CREM + FGCD	84.95	92.86	64.99	81.78	**78.69**	80.65	90.57	0.7416
Baseline + CREM + FGCD + SBE loss	85.53	**93.25**	65.45	81.91	77.67	80.76	90.72	0.7430
Baseline + MCI + CREM + FGCD + SBE loss (IGECNet)	**85.72**	93.19	**66.54**	**82.11**	77.20	**80.95**	**91.33**	**0.7482**

The IoU is aligned with the other evaluation metrics by computing it as

$$\text{IoU} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}}{\text{FN}+\text{FP}+\text{TP}} \tag{24}$$

where TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively, and C is the total number of semantic classes.

The OA quantifies the proportion of correctly classified pixels (true positives and true negatives) across the dataset. It provides a coarse yet intuitive measure of general classification performance. The $F1$-score, which balances precision and recall by computing the harmonic mean, is particularly informative in class-imbalanced scenarios. Precision and recall are, respectively, defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{25}$$

Precision indicates the proportion of correctly predicted positives among all predicted positives, whereas recall is the proportion of true positives recovered among all actual positives. The mean IoU (mIoU) is a class-wise averaged metric that measures the overlap between predicted and GT segmentation regions. Penalizing both over-segmentation and under-segmentation, the mIoU suits multiclass semantic segmentation.

These metrics complement one another and jointly offer a rigorous, comprehensive evaluation of segmentation accuracy, balance, and consistency. Higher values of these metrics indicate stronger generalization and finer discrimination by the model.

C. Implementation Details

The experiments are conducted on a server featuring a 13th-generation Intel[1] Core[2] i9-13900K CPU and an NVIDIA GeForce RTX 4090 24G GPU running Linux version Ubuntu 20.04 LTS. All compared methods are executed using their publicly available codes within the same testing framework and environment, ensuring fair, consistent evaluation. Optimal hyperparameters are selected for each method. The learning rate is adjusted using the Adam optimizer at an initial learning rate of 0.001 and a weight decay of $2.5e^{-4}$. Cosine annealing with warmup and restart is implemented, with the hyperparameters for the first restart iteration and the restart factor set to 15 and 2, respectively. In addition, a batch size of 8 is used across 300 epochs.

In (21), the weights λ, β, and θ for the SBE loss are determined in two steps. First, the magnitudes of the cross-entropy, edge, and elevation losses are examined on the training set, and the initial values are set such that the contributions of different terms are on a comparable scale. Second, empirical tuning within one order of magnitude is performed, with the final values selected based on the best validation performance in terms of mIoU and $F1$ score ($\lambda = 0.1$, $\beta = 10$, and $\theta = 10$). This ensures that each loss component contributes positively while maintaining a balanced optimization.

D. Ablation Studies

1) Ablation Studies of Components: For the Vaihingen dataset, for example, an ablation study is conducted to evaluate the impact of various fusion and feature enhancement techniques on model performance. "Baseline" is the fundamental model (without additional modules), serving as the reference for performance comparison. "Baseline + MCI" incorporates the MCI and is used to examine the effectiveness of its dual-fusion mechanism in improving feature extraction across the HRRSIs and DSMs branches. "Baseline + CREM + FGCD," which integrates the CREM and FGCD, is used to evaluate their combined impact on feature enhancement. "Baseline + CREM + FGCD + SBE loss" further adds the SBE loss and is used to investigate its role in enhancing boundary delineation and enforcing terrain consistency. Finally, "Baseline + MCI + CREM + FGCD + SBE loss (IGECNet)" represents the full model, integrating all proposed components. This configuration demonstrates the cumulative effects of all designed modules on segmentation performance, highlighting the advantages of jointly optimizing feature fusion.

As shown in Table I and Fig. 5, the baseline model has a 77.22% mIoU and a 0.7230 $F1$-score. The IoU values are 91.16% for building, 63.26% for low. veg., 80.78% for tree, and 67.96% for car. This baseline provides a strong foundation by combining features from two layers, but leaves room for refinement in challenging classes, particularly low. veg. and car. MCI integration improves the model's ability to capture both global context and fine-grained details. The results show an increased mIoU of 79.91% and an $F1$-score of 0.7376, with substantial gains in the class-specific IoUs, particularly for building (93.06%) and low. veg. (64.46%). These improvements highlight the MCI's effectiveness in bridging the gap between coarse and detailed feature representations,

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[2]Trademarked.

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TABLE II
ABLATION EXPERIMENTS ON PROPOSED MODULES ON POTSDAM DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA

Methods	IoU/(%)					mIoU/(%)	OA/(%)	F1
	Imp. Suf.	Building	Low. Veg.	Tree	Car			
Baseline	76.79	88.93	67.66	68.41	80.64	76.49	85.48	0.8065
Baseline + MCI	76.75	90.87	64.22	68.11	85.11	77.01	85.16	0.8012
Baseline + CREM + FGCD	78.55	**91.04**	67.54	69.50	85.16	78.36	86.05	0.8118
Baseline + CREM + FGCD + SBE loss	79.06	90.63	67.74	70.08	87.05	78.91	86.35	**0.8253**
IGECNet	**79.13**	90.74	**68.21**	**72.53**	**87.54**	**79.71**	**86.53**	0.8143

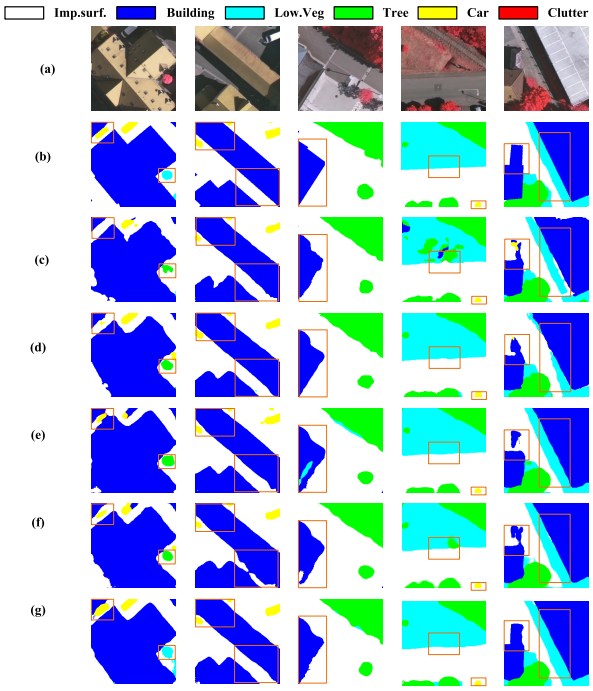


Fig. 5. Visualization of ablation study results. The orange boxes highlight the regions with the most noticeable differences, illustrating the impact of each module on segmentation performance. (a) Image. (b) GT. (c) Baseline. (d) Baseline + MCI. (e) Baseline + CREM + FGCD. (f) Baseline + CREM + FGCD + SBE loss. (g) Baseline + CREM + FGCD + SBE Loss (IGECNet).

thereby improving segmentation across diverse categories. The addition of the CREM and FGCD further enhances segmentation by refining the feature maps and improving consistency across feature gradients. This combination achieves an mIoU of 80.76%, with noticeable gains in challenging classes, such as car and tree. The $F1$-score increases to 0.7416, validating how these modules improve class-specific accuracy and overall segmentation. The addition of the robust SBE loss function into the network further refines the model by enhancing boundary segmentation and addressing class imbalance. This loss function improves the mIoU (80.95%) and $F1$-score (0.7482), especially in challenging scenarios requiring precise segmentation.

For the Potsdam dataset (Table II), a similar trend can be observed. The baseline achieves an mIoU of 76.49% and an $F1$-score of 0.8065. The integration of the MCI leads to modest improvements, particularly in the Building class (from 88.93% to 90.87%), demonstrating its ability to capture cross-modal structural cues. When the CREM and FGCD are introduced, the mIoU increases to 78.36%, with an enhanced IoU for impervious surface (78.55% versus 76.79%) and tree (69.50% versus 68.41%), highlighting the modules' effectiveness in improving feature gradients. Incorporating the SBE loss further boosts the $F1$-score to 0.8253, confirming its contribution to precise boundary delineation and improved terrain consistency.

On the DroneDeploy dataset, shown as Table III, the baseline reaches an mIoU of 84.69% and an $F1$-score of 0.9106. Adding the MCI significantly improves the mIoU to 88.25% and the $F1$-score to 0.9340, with notable gains in the clutter class (61.30%–68.75%) and car class (65.60%–75.34%), which are often challenging due to spectral variability. Further integration of CREM and FGCD raises the mIoU to 89.37% and the $F1$-score to 0.9410, demonstrating consistent benefits across both large-scale and fine-grained categories. With the inclusion of the SBE loss, the network achieves stronger boundary precision, yielding an mIoU of 90.01% and an $F1$-score of 0.9449. Finally, the full IGECNet attains the best overall performance, with an mIoU of 91.55% and an $F1$-score of 0.9605, along with substantial improvements for challenging classes such as clutter (81.26%) and car (85.29%).

According to the above results, although the baseline (which has two-stage feature fusion) provides a solid starting point, the incorporation of the proposed advanced modules (MCI, CREM, and FGCD) consistently boosts performance across all three datasets. The module-wise improvements are not only evident on Vaihingen but also statistically reliable on larger-scale benchmarks such as Potsdam and DroneDeploy, where consistent gains of 2%–6% mIoU are achieved in challenging classes. The combination of all modules produces the optimal network structure (IGECNet), which exhibits superior performance across datasets, thereby confirming both the robustness and the generalization ability of the proposed design.

To further provide intuitive evidence for the module's effectiveness, we visualize the intermediate attention maps of the MCI and the spectral responses of FGCD. As shown in Fig. 6, the model without MCI produces weaker activations along object boundaries and semantically relevant regions. With MCI enabled, cross-attention highlights contours and category-specific cues, confirming that cross-modal interactions improve complementary feature exchange and enhance discrimination. Similarly, Fig. 7 illustrates that FGCD emphasizes mid- to high-frequency components that are strongly correlated with fine-grained structures such as cars and vegetation boundaries. Compared with diffuse activations in the baseline, FGCD suppresses redundant low-frequency noise and

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TABLE III
ABLATION EXPERIMENTS ON PROPOSED MODULES ON DRONEDEPLOY DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA

Methods	IoU/(%)						mIoU/(%)	OA/(%)	F1
	Building	Clutter	Vegetation	Water	Ground	Car			
Baseline	92.67	61.30	88.64	91.25	94.25	65.60	84.69	97.18	0.9106
Baseline + MCI	94.59	68.75	91.28	93.08	95.62	75.34	88.25	97.84	0.9340
Baseline + CREM + FGCD	94.65	71.61	92.34	94.60	95.88	77.58	89.37	97.98	0.9410
Baseline + CREM + FGCD + SBE loss	95.23	73.07	92.89	94.38	96.25	79.09	90.01	98.17	0.9449
IGECNet	**95.76**	**81.26**	**94.60**	**95.19**	**97.24**	**85.29**	**91.55**	**98.55**	**0.9605**

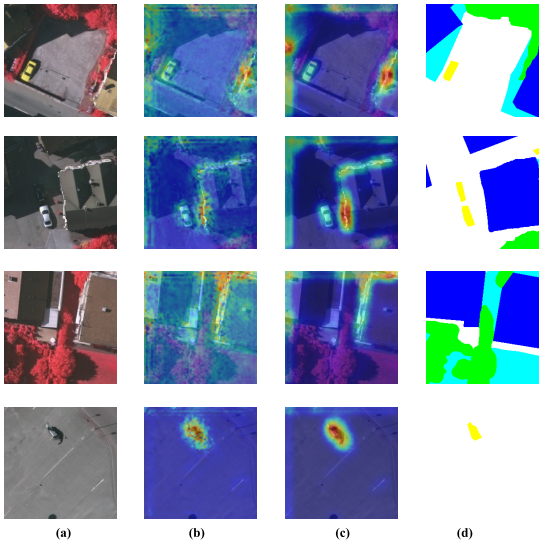


Fig. 6. Visualization of MCI cross-attention responses. (a) Image. (b) w/o MCI. (c) w MCI. (d) GT.

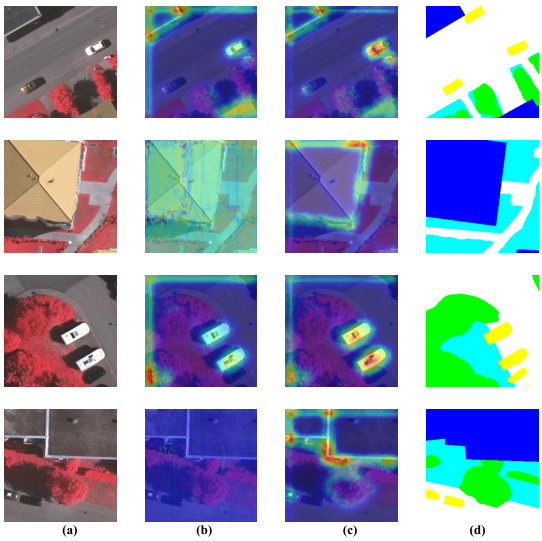


Fig. 7. Visualization of FGCD spectral responses. The module enhances mid- to high-frequency bands that correspond to fine-grained structures and object boundaries. (a) Image. (b) w/o FFT. (c) w FFT. (d) GT.

strengthens boundary localization. These observations validate the mathematical formulations with direct visual evidence, showing how MCI and FGCD enhance cross-modal fusion and frequency domain representation.

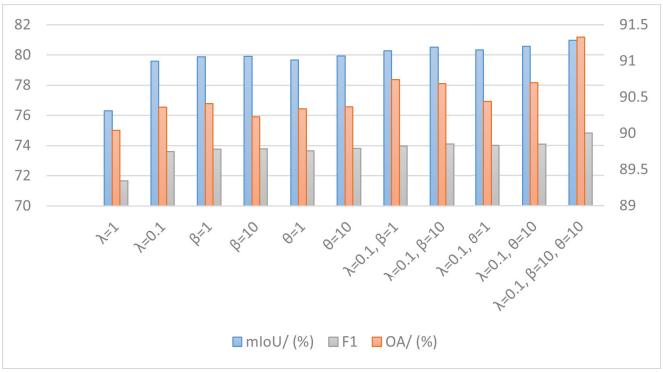


Fig. 8. Results of the combined loss function with respect to the weighting factors λ (midCE), β (Edge), and θ (Elev).

2) Sensitivity Analysis of Loss Function Hyper-Parameters: To further illustrate the role of each component in the combined loss [see (21)], we conduct a sensitivity analysis on the weighting factors λ, β, and θ. The experimental response surface is visualized in Fig. 8. It can be observed that the mIoU improves substantially when all three components are incorporated with moderate weights. In particular, assigning nontrivial weights to the mid-level cross-entropy (λ) and the elevation-guided loss (θ) enhances class balance and terrain awareness, while the edge-preserving term (β) strengthens boundary delineation. Extreme values of a single weight, however, tend to destabilize optimization and reduce accuracy. These findings confirm that the proposed loss function benefits from the complementary effects of its three components, thereby justifying the final choice of weighting factors in our design.

E. Comparison With State-of-the-Art Methods

We comprehensively assess the effectiveness of IGECNet by comparing it with representative methods that implement different multimodal fusion strategies. Baselines encompassing data-, decision-, and feature-level fusion approaches are selected to ensure a thorough analysis.

Data-level fusion methods (e.g., SepNets [49] and RS3Mamba [50]) perform early stage integration by concatenating multimodal inputs. Decision-level fusion methods (e.g., HDenseFormer [51]) aggregate predictions from independent models using ensemble techniques. Feature-level fusion methods, which are further categorized into single- and two-stage feature fusion approaches, integrate multimodal information within the network. Single-stage feature fusion

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TABLE IV

COMPARISON WITH STATE-OF-THE-ART METHODS ON VAIHINGEN DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA, AND THE UNDERLINED ONES ARE THE SECOND BEST

Methods	IoU/(%)					mIoU/(%)	OA/(%)	F1
	Imp. Suf.	Building	Low. Veg.	Tree	Car			
SepGNets [49]	73.15	88.25	60.54	67.19	80.00	73.83	83.39	0.7773
RS3Mamba [50]	82.61	91.10	63.59	81.23	73.03	78.31	89.57	0.7294
HDenseFormer [51]	81.23	88.91	63.99	81.95	70.24	77.27	89.11	0.7343
CMFNet [52]	83.05	90.73	64.39	81.48	78.63	79.66	89.75	0.7368
PACSCNet [29]	83.53	91.13	65.24	82.33	76.31	79.71	90.11	0.7370
AMSFNet [53]	82.96	91.36	62.17	80.49	59.22	75.24	89.33	0.7107
MGFNet [54]	**86.25**	93.12	65.77	81.99	76.64	80.75	90.94	0.7406
GobletNet [55]	84.75	92.72	64.69	81.66	**79.92**	80.75	90.36	0.7428
CFFormer [56]	83.75	91.82	61.37	80.71	67.79	77.09	89.60	0.7218
Ours (IGECNet)	85.72	**93.19**	**66.54**	82.11	77.20	**80.95**	**91.33**	**0.7482**

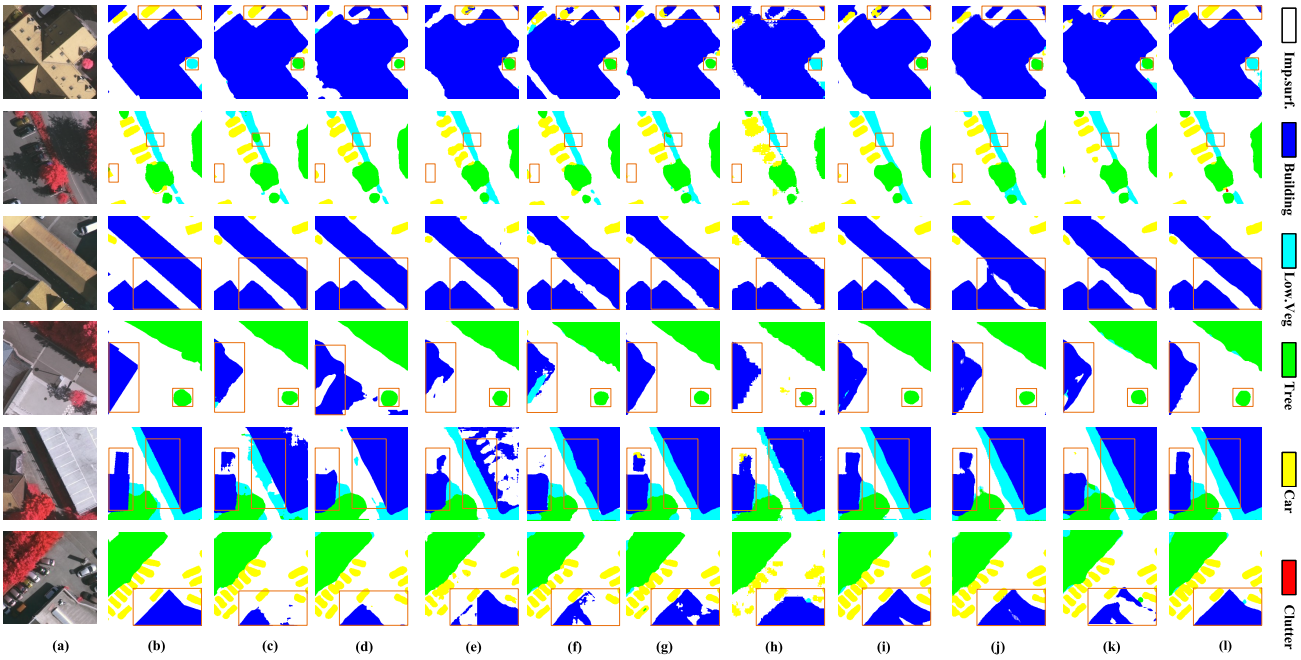


Fig. 9. Visualization results on Vaihingen dataset. The orange boxes highlight the regions with the most noticeable differences. (a) Image. (b) GT. (c) SepGNets (OA=63.46%). (d) RS3Mamba (OA=72.84). (e) HDenseFormer (OA=75.80). (f) CMFNet (OA=81.38). (g) PACSCNet (OA=71.67). (h) AMSFNet (OA=71.76). (i) MGFNet (OA=83.03). (j) GobletNet (OA=77.78). (k) CFFormer (OA=75.80). (l) Ours (IGECNet) (OA=84.83).

approaches (e.g., CMFNet [52], PACSCNet [29], AMSFNet [53], and MGFNet [54]) incorporate multimodal interactions in either the encoder or decoder stage. Two-stage feature fusion methods (e.g., GobletNet [55] and CFFormer [56]) refine multimodal representations across both stages.

To ensure fairness, we clarify that none of the compared methods, including our IGECNet, employ ImageNet-1k or other external pretrained weights. All networks are trained from scratch under identical experimental settings, thereby ensuring that the reported results reflect the intrinsic capacity of each architecture without bias from pretraining. By including methods from different fusion strategy categories in the comparison, we aim to highlight the advantages of IGECNet's two-stage multimodal fusion strategy. Through extensive tests, we demonstrate that IGECNet effectively enhances feature representation, segmentation accuracy, and preservation of object boundaries in HRRSI.

1) Results on ISPRS Vaihingen: The quantitative and qualitative segmentation results on the Vaihingen dataset are presented in Table IV and Fig. 9, respectively. Data-level fusion methods, such as SepGNets and RS3Mamba, typically incorporate ancillary modalities as additional input channels or perform early fusion at the input stage. While these approaches are straightforward and computationally efficient, they lack explicit feature interaction mechanisms, limiting their ability to exploit complementary modality-specific cues. For instance, RS3Mamba, despite leveraging the Mamba architecture to model long-range dependencies, conducts modality fusion merely at the input level. Consequently, its potential to capture cross-modal semantics remains underutilized, resulting in a relatively modest mIoU of 78.31%. Decision-level fusion methods aggregate predictions from modality-specific subnetworks in a late-stage ensemble fashion. Their performance heavily depends on the robustness of individual classifiers and

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TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS ON POTSDAM DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA, AND THE UNDERLINED ONES ARE THE SECOND BEST

Methods	IoU/(%)					mIoU/(%)	OA/(%)	F1
	Imp. Suf.	Building	Low. Veg.	Tree	Car			
SepGNets [49]	73.15	88.25	60.54	67.19	80.00	73.83	83.39	0.7773
RS3Mamba [50]	77.09	87.17	64.74	65.01	84.00	75.60	84.34	0.7913
HDenseFormer [51]	77.54	91.06	64.84	63.15	83.18	75.95	84.67	0.7841
CMFNet [52]	79.00	**91.41**	66.69	72.36	87.05	79.30	86.28	0.8142
PACSCNet [29]	75.08	88.55	63.23	68.73	82.23	75.56	84.20	0.7889
AMSFNet [53]	77.37	90.90	64.49	66.20	80.10	75.81	84.89	0.7937
MGFNet [54]	78.81	90.80	66.56	**72.75**	87.38	79.26	86.29	**0.8186**
GobletNet [55]	78.78	90.90	66.95	72.41	88.13	79.44	86.27	0.8078
CFFormer [56]	78.96	91.02	63.33	65.58	83.80	76.54	85.15	0.7971
Ours (IGECNet)	**79.13**	90.74	**68.21**	72.53	87.45	**79.71**	**86.53**	0.8143

the efficacy of the fusion strategy. However, they typically fail to model intermodal interactions during feature learning. As an example, HDenseFormer yields an mIoU of 77.27%, inferior to most feature-level fusion methods, due to its limited capacity to capture fine-grained modality correlations.

Feature-level fusion methods, which integrate multimodal features within the network architecture, offer more expressive fusion capabilities. Single-stage fusion methods, such as CMFNet and PACSCNet, fuse features at a fixed point (e.g., encoder output), thereby enhancing modality complementation. Nonetheless, their one-shot fusion schemes constrain the network's ability to iteratively refine intermodal dependencies. As a result, their performance (e.g., 79.66% and 79.71% mIoU, respectively) is generally higher than that of data-/decision-level fusion, but still limited in capturing hierarchical multimodal semantics. Two-stage feature fusion approaches, such as GobletNet and CFFormer, refine multimodal feature integration via iterative fusion across both encoder and decoder stages. GobletNet achieves an mIoU of 80.75% through recursive feature aggregation, while CFFormer (mIoU = 77.09%) adopts a cross-scale transformer for multilevel modality interaction. Although CFFormer introduces a strong transformer backbone, its fixed fusion positions and high computational overhead may hinder fine-grained spatial localization in some categories. In contrast, our proposed IGECNet achieves superior performance by explicitly modeling entropy-, frequency-, and cross-modal priors through four complementary modules. These modules enable hierarchical cross-scale and semantic interactions while preserving spatial detail and contextual integrity. As a result, IGECNet not only outperforms traditional fusion paradigms across all three fusion levels but also delivers a balanced improvement across both object-level accuracy and scene-level consistency.

The qualitative comparison findings (Fig. 9) further support the quantitative results. The IGECNet-generated segmentation maps exhibit superior boundary delineation and category consistency, particularly in regions where the other methods struggle. The yellow regions in Fig. 9 are areas where IGECNet outperforms the other methods in segmentation accuracy. Notably, our method effectively preserves fine-grained structures, such as small-scale objects and thin boundaries, which are often overlooked by conventional methods. Hence, our approach robustly handles complex remote sensing images and offers advantages over existing feature fusion strategies. Overall, the experimental results confirm that IGECNet significantly improves multimodal segmentation accuracy by leveraging complementary information from different modalities.

2) Results on ISPRS Potsdam: The quantitative and qualitative segmentation results on the Potsdam dataset are presented in Table V and Fig. 10, respectively. The trends are similar to those on the first dataset. The data-level fusion methods (SepNets and RS3Mamba) perform early stage integration but do not fully exploit complementary multimodal information, leading to suboptimal segmentation, particularly in complex categories (e.g., low. veg. and car). Their mIoU values are below 76%, highlighting the limitation of simple data concatenation. The decision-level fusion method HDenseFormer aggregates predictions from separate models. However, the effectiveness of this late fusion is constrained by the robustness of individual classifiers, resulting in only a marginal improvement (75.95% mIoU). Hence, decision fusion alone is insufficient for effectively integrating multimodal features.

Feature-level fusion methods perform excellently by combining multimodal representations within the network. Single-stage fusion methods improve segmentation by incorporating multimodal cues in either the encoder or the decoder. However, limiting the interaction to a single stage results in incomplete feature utilization, leading to challenges in segmenting fine-scale structures. Two-stage feature fusion methods further improve segmentation accuracy by utilizing fused multimodal features in both the encoder and decoder stages. This comprehensive strategy enhances feature propagation and class distinction, as reflected in mIoU values exceeding 79% for GobletNet. The proposed IGECNet achieves the best overall performance (79.71% mIoU), particularly for low. veg. (68.21% IoU). This is attributed to its dynamic cross-modal refinement (MCI), multiscale integration (CREM and FGCD), and improved boundary preservation (SBE loss). IGECNet produces clearer object boundaries and higher segmentation consistency than other methods, particularly in the challenging regions where low. veg. and tree are not easily distinguished (Fig. 10). These results further validate the effectiveness of our two-stage multimodal fusion strategy and proposed modules in improving segmentation precision.

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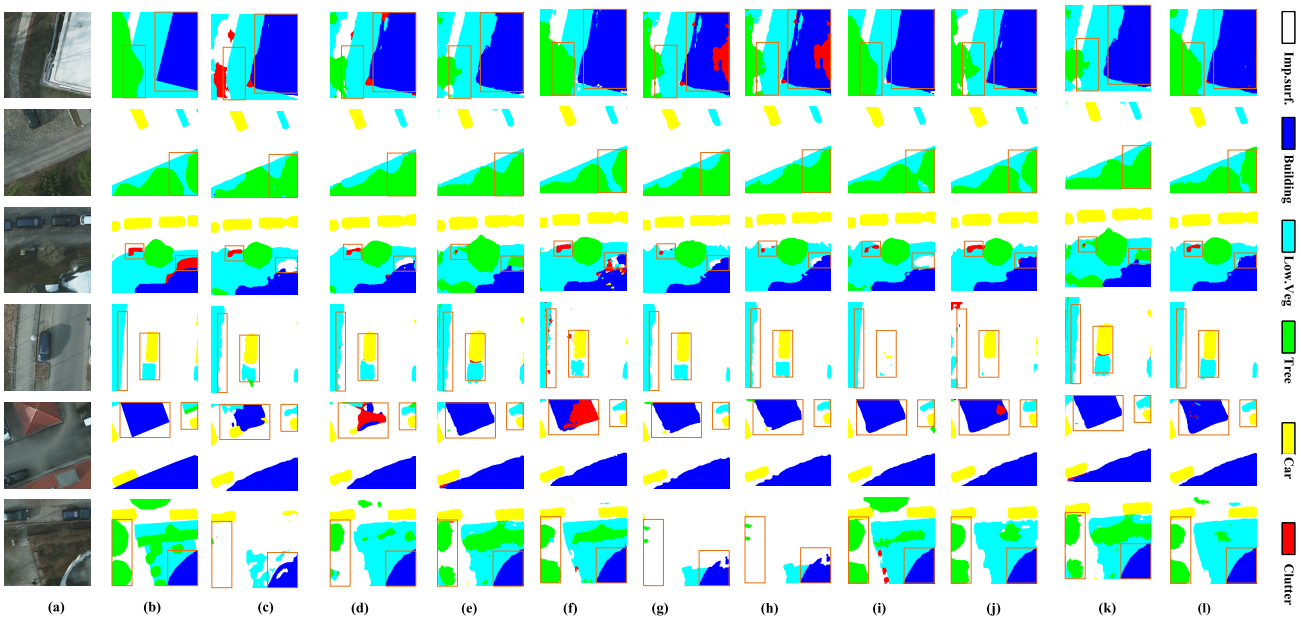


Fig. 10. Visualization results on Potsdam dataset. The orange boxes highlight the regions with the most noticeable differences. (a) Image. (b) GT. (c) SepGNets (OA=92.87). (d) RS3Mamba (OA=92.87). (e) HDenseFormer (OA=89.96). (f) CMFNet (OA=90.77). (g) PACSCNet (OA=91.29). (h) AMSFNet (OA=91.18). (i) MGFNet (OA=94.56). (j) GobletNet (OA=92.93). (k) CFFormer (OA=91.26). (l) Ours (IGECNet) (OA=95.22).

TABLE VI

COMPARISON WITH STATE-OF-THE-ART METHODS ON DRONEDEPLOY SEGMENTATION DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA, AND THE UNDERLINED ONES ARE THE SECOND BEST

Methods	IoU/(%)						mIoU/(%)	OA/(%)	F1
	Building	Clutter	Vegetation	Water	Ground	Car			
SepGNets [49]	95.28	78.08	93.55	94.62	96.65	73.56	88.62	98.27	0.9452
RS3Mamba [50]	89.95	57.88	88.27	88.29	93.76	60.00	82.41	96.85	0.8953
HDenseFormer [51]	83.10	40.60	80.64	83.89	89.52	45.95	70.62	94.38	0.8352
CMFNet [52]	**95.83**	76.13	93.33	94.74	95.91	79.90	90.58	97.99	0.9486
PACSCNet [29]	92.81	70.65	91.31	92.54	95.51	73.57	87.80	97.60	0.9317
AMSFNet [53]	89.34	47.18	85.43	90.01	92.24	49.41	75.60	96.15	0.8669
MGFNet [54]	95.66	76.62	94.24	94.47	96.81	76.80	89.10	98.39	0.9478
GobletNet [55]	95.44	74.26	92.66	90.25	95.47	81.09	88.19	97.72	0.9429
CFFormer [56]	92.79	67.26	90.51	93.35	94.75	77.67	87.82	97.40	0.9317
Ours (IGECNet)	95.76	**81.26**	**94.60**	**95.19**	**97.24**	**85.29**	**91.55**	**98.55**	**0.9605**

TABLE VII

BACKBONE SELECTION EXPERIMENTS ON VAIHINGEN DATASET

Backbone	Params (M)	FLOPs (G)	mIoU (%)
Xception	484.04	146.21	78.20
MobileNet v3 (small)	9.34	26.81	75.24
ResNet50	439.22	220.90	78.88
ResNeXt50	104.74	2421.96	79.05
MPViT	753.40	599.45	80.76
MixViT	72.07	52.96	75.44
MambaVision	368.59	238.00	79.03
Res2Net50	1916.52	316.66	**80.95**

3) Results on DroneDeploy Segmentation: The quantitative and qualitative segmentation results on the DroneDeploy Segmentation dataset are presented in Table VI and Fig. 11, respectively. As for the previous datasets, the data-level fusion methods SepNets and RS3Mamba perform early stage integration through direct multimodal concatenation and thus have limited ability to exploit cross-modal complementarity fully, leading to suboptimal segmentation, particularly for clutter and car. Their mIoU values are below 80%, indicating that simple input concatenation is insufficient for capturing complex scene variations. The decision-level fusion method HDenseFormer aggregates predictions from independent models, but the effectiveness of this late fusion approach is constrained by the individual robustness of classifiers. Its mIoU (70.62%) indicates that decision fusion alone is insufficient for fully integrating multimodal features, resulting in performance degradation, particularly for small-scale objects with high spectral–spatial variability. The feature-level fusion methods perform well by integrating multimodal representations within the network. The single-stage fusion approaches CMFNet, PACSCNet, and AMSFNet improve segmentation accuracy by incorporating multimodal interactions in either the encoder or decoder stage. However, their inability to propagate fused features across both stages results in incomplete feature utilization, particularly for small-scale structures (e.g., cars and clutter), where their segmentation performance is

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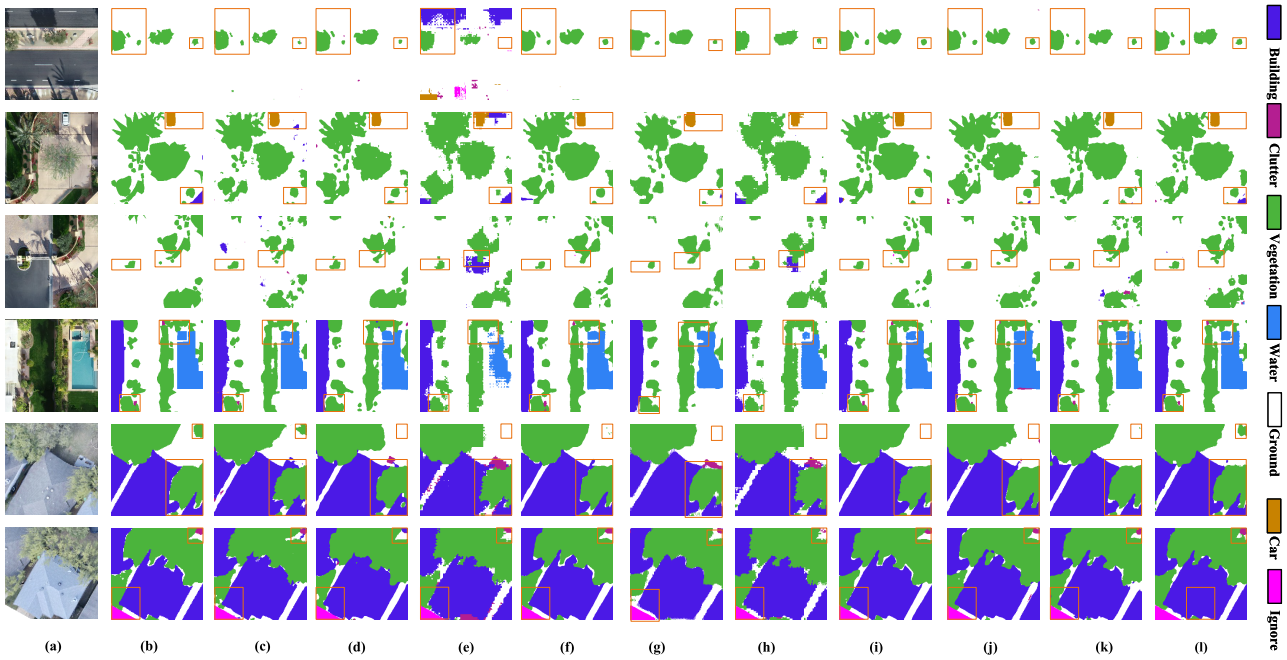


Fig. 11. Visualization results on DroneDeploy Segmentation dataset. The orange boxes highlight the regions with the most noticeable differences. (a) Image. (b) GT. (c) SepGNets (OA=96.48). (d) RS3Mamba (OA=97.01). (e) HDenseFormer (OA=81.66). (f) CMFNet (OA=97.40). (g) PACSCNet (OA=96.99). (h) AMSFNet (OA=96.61). (i) MGFNet (OA=97.50). (j) GobletNet (OA=97.15). (k) CFFormer (OA=97.34). (l) Ours (IGECNet) (OA=98.45).

limited. The two-stage feature fusion methods GobletNet and CEGFNet enhance segmentation accuracy by refining multimodal representations across both the encoder and decoder stages, thereby improving feature propagation and interclass distinction. However, the proposed IGECNet achieves the best overall performance (91.55% mIoU), particularly for car (85.29% IoU). This improvement is ascribed to the MCI mechanism for dynamic cross-modal refinement, the CREM and FGCD modules for multiscale feature integration, and the SBE loss for improved boundary preservation.

IGECNet produces clearer object boundaries and more consistent segmentation results than the other methods, especially in complex regions where clutter and vegetation overlap (Fig. 11). These findings further validate the effectiveness of our two-stage multimodal fusion strategy and the proposed modules in improving segmentation precision for HRRSIs.

4) Model Efficiency: To ensure that the proposed framework benefits from both representation power and computational feasibility, we first evaluated several popular backbone networks, as summarized in Table VII. The comparison covers lightweight architectures (MobileNet v3 [57]), conventional convolutional backbones (ResNet50 [58], ResNeXt50 [59]), more advanced designs (Xception [60] and Res2Net50 [61]), and recently proposed global-modeling design (MPViT [62], MixViT [63] and MambaVision [64]). To ensure transparency and reproducibility, all parameter and FLOPs measurements in this study were computed under a consistent tensor configuration, using a batch size of 1 with input resolutions of $1 \times 3 \times 256 \times 256$ for HRRSI and $1 \times 1 \times 256 \times 256$ for DSM, respectively.

From the results, MobileNet v3 offers high efficiency but limited accuracy due to its constrained capacity. Conventional backbones such as ResNet50 and Xception deliver moderate performance, yet still fall short in capturing fine-grained spatial-spectral details in complex scenes. ResNeXt50 achieves competitive accuracy, but its extremely high computational cost (over 2400-G FLOPs) hinders practical use. When incorporating global-modeling backbones like MPViT, MixViT, and MambaVision under the same training settings and without large-scale pretraining, MPViT attains comparable performance at a much higher computational cost (599.45-G FLOPs), while MixViT and MambaVision show slightly lower performance with modestly reduced FLOPs. In contrast, Res2Net50 offers a more favorable tradeoff, achieving the highest mIoU (80.95%) with a moderate FLOP budget.

In addition, to further provide an intuitive understanding of the tradeoff between representational richness and computational feasibility, we visualize Params, FLOPs, and mIoU simultaneously of compared methods and the proposed IGECNet in Fig. 12. It can be observed that IGECNet consistently achieves the highest mIoU on all three datasets, while its FLOPs remain considerably lower than other high-capacity feature fusion models such as CMFNet and GobletNet. This 3-D perspective highlights that IGECNet attains a favorable balance between accuracy and efficiency, thereby supporting its applicability in practical multimodal remote sensing tasks. From the results, we can observe several important trends. First, lightweight models such as SepNets and HDenseFormer achieve competitive efficiency but suffer from limited segmentation accuracy due to insufficient modeling capacity and shallow feature interactions. On the other hand, transformer-based methods such as MambaFormer exhibit relatively low FLOPs but rely heavily on long-sequence modeling, resulting

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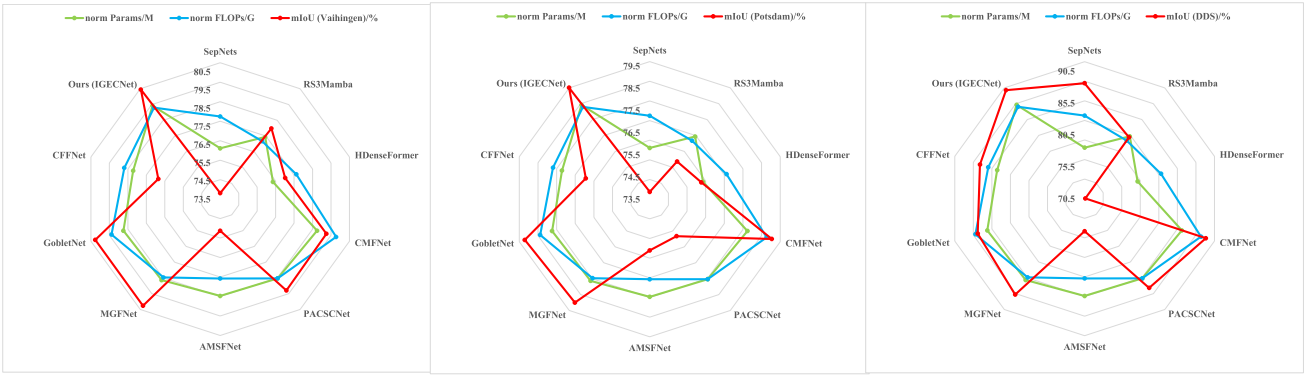


Fig. 12. Radar-chart comparison of efficiency-performance characteristics across three benchmark datasets. These charts illustrate the relationships among model capacity (Params), computational cost (FLOPs), and segmentation results (mIoU). A unified logarithmic and normalized scaling is applied to the data to ensure performance visualization. IGECNet achieves the highest mIoU across all three datasets while maintaining computational efficiency within a reasonable range compared with other large-capacity fusion models.

TABLE VIII
COMPARISON OF DIFFERENT FUSION STRATEGIES ON VAIHINGEN DATASET. THE BOLDFACE NUMBERS ARE THE BEST DATA

Methods	IoU/(%)					mIoU/(%)	OA/(%)	F1
	Imp. Suf.	Building	Low. Veg.	Tree	Car			
Data-level fusion	81.61	89.58	62.15	80.59	70.23	76.83	88.85	0.7213
Decision-level fusion	82.50	90.24	63.54	79.39	68.28	76.79	89.03	0.7211
Single-stage feature fusion-decoder	79.16	91.05	**68.75**	75.33	**70.43**	76.94	87.38	**0.7662**
Single-stage feature fusion-encoder	81.63	89.49	62.60	80.52	69.69	76.79	88.85	0.7211
Two-stage feature fusion	**82.94**	**91.16**	63.26	**80.78**	67.96	**77.22**	**89.50**	0.7231

in weaker fine-grained localization capability. Feature-level fusion networks (e.g., CMFNet, PACSCNet, MGFNet, and GobletNet) introduce substantial parameter and computational overhead. For instance, CMFNet contains nearly 788-G FLOPs, while GobletNet reaches 369.2-G FLOPs. These heavy architectures provide improved accuracy but are computationally prohibitive for large-scale deployment. CFFormer strikes a moderate balance, yet still incurs non-negligible computational costs. In contrast, our proposed IGECNet is parameter-heavy due to the introduction of multibranch modules and extensive feature refinement. Nevertheless, it maintains relatively favorable FLOPs (316.66 G) compared with other high-capacity fusion networks such as CMFNet and GobletNet. This indicates that IGECNet achieves superior segmentation accuracy while preserving computational feasibility at the cost of increased parameter storage. The efficiency results highlight that IGECNet prioritizes representational richness and fine-grained multimodal interaction, trading off model size for a better balance between accuracy and computational complexity.

IV. DISCUSSION

This section discusses the implications of our proposed methodology, focusing on two key aspects. The first focus is the effectiveness, specifically the advantages and limitations, of different fusion strategies (data-level, decision-level, and feature fusion approaches). The second is the efficiency limitations of IGECNet, particularly due to the additional computational overhead introduced by image entropy and clustering-based processing. In addition, potential research directions for improvement are explored to enhance model adaptability and performance in complex real-world scenarios.

A. Analysis of Fusion Strategies

We analyze the performance of the compared fusion strategies at varying stages: data-level, single-stage feature, and decision-level fusions. Strategic performance varies depending on the level where modality integration occurs, as shown in Table VIII.

Data-level fusion, which directly concatenates input data, achieves an mIoU of 76.83% and an $F1$-score of 0.7213. Although it effectively retains low-level spatial information, its inability to model complex feature dependencies leads to suboptimal results for challenging classes, such as low. veg. decision-level fusion introduces the learnable parameters λ and β for dynamically adjusting the contributions of different modalities in the prediction stage. This strategy achieves an mIoU of 76.79% and an $F1$-score of 0.7211. However, because it operates on the final prediction outputs without directly improving feature representation, it is less effective in refining fine-grained details. Feature-level fusion is more effective. Single-stage feature fusion, which integrates multimodal features in either the encoder or the decoder, enhances segmentation by capturing both low- and high-level representations. The decoder-based variant achieves an mIoU of 76.94% and an $F1$-score of 0.7662, particularly improving segmentation accuracy for Low. veg. and car. By contrast, the encoder-based approach yields an mIoU of 76.79%. Further improvements are observed under two-stage feature fusion, which aggregates features from multiple layers. This

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TABLE IX
ROBUSTNESS ANALYSIS OF IGECNET UNDER IMPERFECT DSMS CONDITIONS

Setting	Imp. Surf.	Building	Low. Veg.	Tree	Car	mIoU	OA	F1
Random missing 10%	83.63	88.64	63.64	80.28	71.47	77.53	89.15	72.55
Random missing 20%	79.94	85.71	62.45	78.81	65.04	74.39	87.64	70.84
Random missing 50%	77.74	78.75	56.54	71.05	61.98	69.21	83.33	67.91
Block missing 8	85.12	90.72	64.03	81.52	70.96	78.47	90.09	73.00
Block missing 16	84.49	90.65	64.06	80.93	69.24	77.87	89.83	72.68
Block missing 32	83.37	90.23	64.44	81.17	69.29	77.70	89.69	72.61
Gaussian noise 5%	85.19	91.27	63.94	81.14	66.79	77.67	90.02	72.52
Gaussian noise 10%	84.29	88.48	63.49	79.55	66.12	76.38	89.04	71.87
Gaussian noise 20%	81.93	89.16	61.07	78.67	62.87	74.74	88.26	70.91
Salt-and-pepper 5%	84.58	89.87	63.23	80.74	69.99	77.68	89.57	72.58
Salt-and-pepper 10%	84.00	89.21	62.71	80.54	63.28	75.95	89.22	71.57
Salt-and-pepper 20%	83.19	87.88	62.65	78.49	61.95	74.83	88.49	70.98

TABLE X
OPERATION SUMMARY OF IGECNET COMPONENTS

Eq.	**Input shape**	**Operations**	**Output shape**
Eq. (1)	$F_H \in \mathbb{R}^{256\times64\times64}$	Dilated convolution; concatenate; dilated convolution, BN, SiLU	$F_{H_{DC}}, F_{H_{mul}}, F_{fu_H} \in \mathbb{R}^{256\times64\times64}$
Eq. (2)	$F_D \in \mathbb{R}^{256\times64\times64}$	Same as above	$F_{D_{DC}}, F_{D_{mul}}, F_{fu_D} \in \mathbb{R}^{256\times64\times64}$
Eq. (3)	$F_{fu_D}, F_{fu_H} \in \mathbb{R}^{256\times64\times64}$	Convolution	$Q, K \in \mathbb{R}^{32\times64\times64}$, $V \in \mathbb{R}^{256\times64\times64}$
Eq. (4)	$Q_h \in \mathbb{R}^{64\times64\times32}$, $K_h \in \mathbb{R}^{64\times32\times64}$	Matrix multiplication	$S_h \in \mathbb{R}^{64\times64\times64}$
Eq. (5)	$S_h, S_w \in \mathbb{R}^{64\times64\times64}$	Concatenate, softmax	$\mathrm{Attn} \in \mathbb{R}^{64\times64\times128}$
Eq. (6)	$\mathrm{Attn}_h \in \mathbb{R}^{64\times64\times64}$, $V_h \in \mathbb{R}^{64\times256\times64}$	Matrix multiplication	$CA_h \in \mathbb{R}^{256\times64\times64}$
Eq. (7)	$CA_h, CA_w, F_{fu_D}, F_{fu_H} \in \mathbb{R}^{256\times64\times64}$	Add, Hadamard product; convolution, concatenate, convolution, BN, ReLU	$CA_{mid}, CA_{fu_D}, CA_{fu_H} \in \mathbb{R}^{256\times64\times64}$
Eq. (8)	$E_c, G_c \in \mathbb{R}^{256}$	Tensor stack	$A_c \in \mathbb{R}^{256\times2}$
Eq. (9)	$X \in \mathbb{R}^{256\times64\times64}$	Entropy computation	$E_c \in \mathbb{R}^{256}$
Eq. (10)	$X \in \mathbb{R}^{256\times64\times64}$	Global average pooling	$G_c \in \mathbb{R}^{256}$
Eq. (11)	$A_c \in \mathbb{R}^{256\times2}$	Cluster	C_k (four clusters of channel indices)
Eq. (12)	$X_{C_k} \in \mathbb{R}^{k\times64\times64}$	Convolution	$f_k \in \mathbb{R}^{64\times64\times64}$
Eq. (13)	$f_k \in \mathbb{R}^{64\times64\times64}$	Concatenate	$f_{\mathrm{Enh}} \in \mathbb{R}^{256\times64\times64}$
Eq. (14)	$X \in \mathbb{R}^{128\times256\times256}$	FFT, Hadamard product	$X_f, Y_f \in \mathbb{R}^{256\times129\times128}$
Eq. (15)	$Y_f \in \mathbb{R}^{256\times129\times128}$	IFFT	$Y \in \mathbb{R}^{128\times256\times256}$

method achieves the highest mIoU (77.22%) and an $F1$-score of 0.7231, demonstrating superior performance across most categories.

The inferior performance of the two-stage fusion model on low. veg. and car relative to single-stage decoder-based fusion may be due to excessive feature smoothing. This process weakens fine-grained details, which are crucial for these categories. In addition, deeper fusion layers may suppress high-frequency information, influencing the segmentation of small or texture-sensitive objects. The adaptive weighting of multilevel features may better preserve category-specific details.

B. Robustness Analysis Under Imperfect DSMs Inputs

In real-world multimodal remote sensing applications, DSM data may be incomplete or affected by various types of noise due to acquisition constraints and environmental factors. To investigate the stability of IGECNet under such imperfect conditions, we introduced four types of degradation into the DSMs inputs: 1) random missing values; 2) block missing values; 3) Gaussian noise; and 4) salt-and-pepper noise. The quantitative results are summarized in Table IX.

As shown in Table IX, segmentation performance exhibits a gradual decline as noise intensity or missing ratios increase. Nevertheless, IGECNet consistently maintains an acceptable level of accuracy under all tested conditions. For instance, even with severe degradation such as 50% random missing values, the network achieves 69.21% mIoU and 83.33% OA, while with moderate block occlusions (block size = 16), the mIoU and OA remain at 77.87% and 89.83%, respectively. This robustness can be attributed to the architectural design, where modules such as MCI, CREM, FGCD, and the SSBE loss function enable the network to exploit complementary information from HRRSI. As a result, the negative effects of degraded DSM inputs are alleviated, and the overall performance remains within a practically acceptable range.

In summary, the robustness analysis confirms that IGECNet is capable of handling imperfect DSM inputs and delivering stable performance in scenarios where DSM data are noisy or partially missing, thereby enhancing its practical applicability.

C. Limitations of IGECNet

While the proposed method effectively enhances feature representation, it introduces additional computational overhead

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due to the integration of image entropy and clustering-based processing. The entropy-guided feature refinement process, which involves calculating local and global entropy distributions, increases preprocessing time and the overall complexity of the network. Similarly, the clustering-based module requires iterative optimization steps, which elevate memory usage and inference latency, particularly when applied to HRRSIs data. These factors may limit the real-time applicability and scalability of IGECNet, especially for large-scale or resource-constrained environments. To mitigate these limitations, future research could focus on developing more efficient entropy estimation techniques or adopting approximate clustering methods that reduce computational complexity. In addition, utilizing model optimization strategies, such as knowledge distillation and lightweight network architectures, could accelerate inference while preserving segmentation accuracy. Moreover, hardware-aware optimization techniques, including pruning and quantization, could further improve efficiency without significantly compromising performance.

In addition, the SBE loss is specifically designed to exploit elevation information embedded in DSMs data, which may not directly transfer to other modality combinations such as SAR, LiDAR, or hyperspectral imagery. Nonetheless, the proposed cross-modal interaction and feature guidance strategies are generalizable, and only minor adaptations (e.g., modifying input channels or redefining priors according to the physical characteristics of new modalities) are required to extend the framework. Future work will investigate these extensions to broaden the applicability of IGECNet across diverse multimodal remote sensing scenarios.

V. Conclusion

Herein, we propose IGECNet, a multimodal cross-scale integration and elevation consistency segmentation network designed to effectively integrate complementary information from different modalities through a two-stage feature fusion framework. The network leveraged multimodal correlations by incorporating MCI for enhanced feature fusion, while CREM and FGCD supported multiscale feature integration and spatial awareness, thereby improving segmentation accuracy. In addition, the SBE loss was introduced to enhance boundary delineation and elevation consistency. Extensive experiments on benchmark datasets demonstrated that IGECNet consistently outperformed state-of-the-art methods, confirming the effectiveness of the proposed modules.

Future work will focus on extending the IGECNet framework to multitemporal remote sensing tasks, such as change detection, to better capture temporal dependencies in multimodal data. Furthermore, we plan to investigate weakly supervised and self-supervised learning strategies to reduce reliance on extensive labeled data, thereby enhancing the model's scalability and adaptability in large-scale and diverse remote sensing applications.

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